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## 1 General Strategy

### 1.1 • Before Contest

- Check whether or not `bits/stdc++.h` is working.
- And also `ext/pb_ds`.
- And `__int128`.
- Check how many operations you can run in 1 second.
- Check how many bits long double uses.

### 1.2 • Soft Techniques

- Did you read the problem once more?
- Is your claim actually true?
- Have you tried representing the problem as a formula?
- Can you unify the cost function so that every action have non-negative cost?
- What about just playing around with the constraint, with no intention to reach the goal?
- What about starting from scratch, with completely different ideas?
- Have you examined the small cases?
- What is the range of the potential answer?
- Is there some auxiliary property that I can use?
- Why don't you just get up and start implementing?
- Why don't you just get up and start writing everything?

### 1.3 • Hard Techniques

- Dynamic Programming? Divide and Conquer?
- Greedy? Network Flow? Regret Greedy?
- Square Root Decomposition? Or other bucket size?
- Sparse Table with  $O(1)$  query time?
- Prefix Sum? Or considering  $\leq x$  instead of  $= x$ ?
- Stars and Bars Argument to solve Sum of Product Score?
- Backtracking? Randomization? Heuristics?
- Global Monovariant, with no Locality?

## 2 Code Snippet - C++

### 2.1 • Basic Template

```
#include <bits/stdc++.h>
#define endl '\n'
const int PRECISION = 0;
using namespace std;

void Main(){

}

int main(){
```

```
ios_base::sync_with_stdio(0); cin.tie(0); cout.tie(0);
cout.setf(ios::fixed); cout.precision(PRECISION); Main();
}
```

### 2.2 • Line Input

```
inline void icl(istream& in){ in.ignore(998244353, '\n'); }
// After cin and Before getline
string s; getline(cin, s); // No newline at the end
```

### 2.3 • Vector Manipulation

```
template <typename T> inline void unq(vector<T>& v){
    sort(v.begin(), v.end()); v.erase(unique(v.begin(),
        v.end()), v.end());
}
template <typename T> inline int cvt(vector<T>& v, T x){
    return lower_bound(v.begin(), v.end(), x) - v.begin();
}
template <typename T> inline void erase(vector<T>& v, const
T& x){
    v.erase(remove(v.begin(), v.end(), x), v.end());
} // removes every x in v
```

### 2.4 • Bitwise Function

```
inline int bit1(int x){ return __builtin_popcount(x); } // #
of 1
inline int bit2(int x){ return x==0 ? 32 : __builtin_ctz(x);
} // max k s.t. n | 2^k
inline int bitl(int x){ return x==0 ? 0 : 32 -
__builtin_clz(x); } // # of bits
inline int bith(int x){ return x==0 ? 0 : 1 << bitl(x)-1; }
// max 2^k s.t. 2^k <= n
inline int bitp(int x){
    int y = bith(x); return x==0 ? 0 : y << (x!=y);
} // min 2^k s.t. n <= 2^k
```

### 2.5 • Randomization

```
const time_t TIME =
chrono::high_resolution_clock::now().time_since_epoch().count()
mt19937 gen(TIME);
uniform_int_distribution<int> rng(a, b); // [a, b] range
int value = rng(gen); shuffle(begin, end, gen);
```

### 2.6 • Custom Hash

```
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }

    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
            chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};
```

```
}
};

unordered_map<ll, int, custom_hash> mp;

Custom Hash Table
```

```
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
```

```
gp_hash_table<key, value, hash> table;
```

### 2.7 • Policy Based Data Structure

```
#include <ext/pb_ds/tree_policy.hpp>
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
template <typename T> using ordered_set = tree<T, null_type,
less<T>, rb_tree_tag, tree_order_statistics_node_update>;
template <typename K, typename V> using ordered_map = tree<K,
V, less<K>, rb_tree_tag, tree_order_statistics_node_update>;
```

```
ordered_set<int> s; s.insert(value);
s.find_by_order(index); // 0-based, returns iterator
s.order_of_key(value); // 0-based, lower_bound
```

```
ordered_map<int, int> mp;
mp.insert({1, 4}); mp[4] = 7;
mp.find_by_order(1)->first // key, 4
mp.find_by_order(1)->second // value, 7
```

### 2.8 • Custom Comparison

```
bool cmp(int a, int b){ return a < b; }
sort(begin, end, cmp); // a -> b
set<int, decltype(&cmp)> s(cmp); // a -> b (inorder)
priority_queue<int, vector<int>, decltype(&cmp)> pq(cmp); //
b on top
```

```
sort(v.begin(), v.end(), [](pi2 a, pi2 b){ return a.fr+a.sc <
b.fr+b.sc; });
auto st = lower_bound(v.begin(), v.end(), x, [](pi2 p, int
x){ return p.fr+p.sc < x; });
auto ed = upper_bound(v.begin(), v.end(), x, [](int x, pi2
p){ return x < p.fr+p.sc; });
// [st, ed) is where p.fr+p.sc == x
```

```
struct cmp{
    bool operator()(const pi2& p1, const pi2& p2){ return
        p1.fr+p1.sc < p2.fr+p2.sc; }
};
tree<pi2, null_type, cmp, rb_tree_tag,
tree_order_statistics_node_update> s;
```

### 2.9 • Dynamic Bitset

```
const int MAX = 200000;
template <int N = 1> void solve(int n){
    if (N < n){ return solve<min(N*2, MAX)>(n); }
}
```

### 3 Code Snippet - Python

#### 3.1 Basic Template

```
import sys; input = lambda: sys.stdin.readline().rstrip('\n')
inputs = lambda t, l=tuple: l(map(t, input().split()))
sys.setrecursionlimit(100000)
```

#### 3.2 List Manipulation

```
l.reverse(); l = list(reversed(l))
l.sort(); l = sorted(l)
```

#### 3.3 Arbitrary Precision

```
from fractions import * # Fraction, no limit
x = Fraction(10)
```

```
from decimal import * # Decimal, limited
getcontext().prec = 28 # Number of precision = digits
x = Decimal(20)
```

#### 3.4 Data Structure

```
from collections import deque
q = deque() # Deque
q.append(x); q.appendleft(x)
x = q.pop(); x = q.popleft()
```

```
import heapq # Heap
pq = []
heapq.heappush(pq) # push
x = heapq.heappop(pq) # pop
```

#### 3.5 Custom Hashing

```
import time; TIME = int(time.time() * 1000)
dict[(x, TIME)] # anti-hash
```

#### 3.6 Custom Comparison

```
from functools import cmp_to_key
def cmp(a, b): return sgn(a-b) # a < b then -, a > b then +,
otherwise 0
l.sort(key=cmp_to_key(cmp)); sorted(l, key=cmp_to_key(cmp))
```

### 4 Mathematics - Number Theory

#### 4.1 Floor & Ceil of a Fraction

```
inline ll ffloor(ll a, ll b){
    if (b < 0){ a = -a; b = -b; }
    return (a >= 0 ? a/b : (-a+b-1)/b*-1);
}
inline ll fceil(ll a, ll b){
    if (b < 0){ a = -a; b = -b; }
    return (a >= 0 ? (a+b-1)/b : -a/b*-1);
}
```

#### 4.2 Lifting the Exponent Lemma

Let  $p$  be a prime, and  $x$  and  $y$  be integers. Let  $v_p(x)$  be the number of times you can divide  $x$  by  $p$ .

- If  $p$  is odd:
  - If  $p \mid x - y$ , then  $v_p(x^n - y^n) = v_p(x - y) + v_p(n)$ .
  - If  $p \mid x + y$  and  $n$  is odd, then  $v_p(x^n + y^n) = v_p(x + y) + v_p(n)$ .
  - If  $p \mid x + y$  and  $n$  is even, then  $v_p(x^n + y^n) = 0$ .
- If  $p = 2$ :
  - If  $p \mid x - y$  and  $n$  is odd, then  $v_p(x^n - y^n) = v_p(x - y)$ .
  - If  $p \mid x - y$  and  $n$  is even, then  $v_p(x^n - y^n) = v_p(x - y) + v_p(x + y) + v_p(n) - 1 = v_p((x^2 - y^2)/2) + v_p(n)$ .
  - If  $p^2 \mid x - y$  and  $x$  and  $y$  are both odd, then  $v_p(x^n - y^n) = v_p(x - y) + v_p(n)$ .
  - If  $p \mid x + y$  and  $n$  is odd, then  $v_p(x^n + y^n) = v_p(x + y)$ .
  - If  $p \mid x + y$  and  $n$  is even, then  $v_p(x^n + y^n) = 1$ .
- For all  $p$ :
  - If  $p \mid x - y$  and  $p \nmid n$ , then  $v_p(x^n - y^n) = v_p(x - y)$ .
  - If  $p \mid x + y$  and  $p \nmid n$  and  $n$  is odd, then  $v_p(x^n + y^n) = v_p(x + y)$ .

#### 4.3 Extended Euclidean Algorithm

The solutions to  $ax + by = g$  is  $(x + bk/g, y - ak/g)$ .

```
pl3 egcd(ll a, ll b){
    if (b == 0){ return {{1, 0}, a}; }
    pl3 p = egcd(b, a%b);
    ll x = p.fr.fr, y = p.fr.sc, g = p.sc;
    ll xx = y, yy = x - a/b*y;
    return {{xx, yy}, g};
}
```

#### 4.4 Chinese Remainder Theorem

Also check [Extended Euclidean Algorithm](#) at page 3.

```
inline ll finv(ll a, ll b){ pl3 p = egcd(a, b); return
(p.fr.fr%n+n)%n; }
pl2 crt(pl2 f1, pl2 f2){
    if (f1.sc < f2.sc){ swap(f1, f2); }
    ll a1 = f1.fr, m1 = f1.sc; ll a2 = f2.fr, m2 = f2.sc;
    ll g = gcd(m1, m2); ll l = lcm(m1, m2);
    if ((a2-a1)%g != 0){ return {-1, -1}; }
    ll mg1 = m1/g, mg2 = m2/g, ag = (a2-a1)/g;
    ll y = ag * finv(mg1, mg2) %mg2;
    ll x = m1*y + a1; return {(x%l+1)%l, l};
}
```

#### 4.5 Möbius Function

```
int phi[X+20], mob[X+20], prr[X+20];
memset(mob, -1, sizeof(mob)); mob[1] = 1;
for (int x = 1; x <= X; x++){ phi[x] = x; }
for (int x = 2; x <= X; x++){
    if (prr[x] == 0){
```

```
    for (int a = x; a <= X; a+=x){ prr[a] = x; phi[a] -=
        phi[a]/x; }
    } int p = prr[x]; mob[x] = (x/p%p == 0 ? 0 :
        mob[x/p]*mob[p]);
}
```

#### 4.6 Miller-Rabin

```
const vector<ll> prr = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
31, 37};
bool prime(ll n){
    if (n <= 40){
        for (int p : prr){ if (n == p){ return 1; } }
        return 0;
    } if (n%2 == 0){ return 0; }
    int s = 0; ll d = n-1; while (d%2 == 0){ s += 1; d /= 2; }
    for (int p : prr){
        ll res = 1, mul = p, bit = d; while (bit){
            if (bit&1){ res = (i128)res*mul % n; }
            mul = (i128)mul*mul % n; bit >>= 1;
        }
        bool chk = (res == 1); for (int r = 0; r < s; r++){
            chk |= (res == n-1); res = (i128)res*res % n;
        } if (!chk){ return 0; }
    } return 1;
}
```

#### 4.7 Pollard's Rho

```
vector<ll> ans;
inline ll f(ll x, ll c, ll mod){ return ((i128)x*x + c) %
mod; }
void factor(ll n){
    if (n == 1){ return; }
    if (n%2 == 0){ ans.push_back(2); return factor(n/2); }
    if (prime(n)){ ans.push_back(n); return; }
    uniform_int_distribution<ll> rnd(1, n);
    ll x1 = rnd(gen); ll x2 = x1;
    ll c = rnd(gen); do{
        x1 = f(x1, c, n); x2 = f(f(x2, c, n), c, n);
    } while (gcd(abs(x1-x2), n) == 1);
    ll g = gcd(abs(x1-x2), n);
    if (g == n){ return factor(n); }
    else{ factor(g); factor(n/g); }
}
```

#### 4.8 Power Tower

$\text{CUT} \geq \log_2 m$  is that  $a^{\phi(m)} \not\equiv 1 \pmod{m}$  can happen. But for  $d \geq \log_2 m$ ,  $a^{d+\phi(m)} \equiv a^d \pmod{m}$ .

```
const ll INF = 1e18; // INF > mod^2; might need i128
ll fpow(ll mul, ll bit, ll mod, ll lim = INF){
    ll res = 1; while (bit){
        if (bit&1){ res = min(res*mul, lim) % mod; }
        mul = mul*mul % mod; bit >>= 1;
    } return res;
}
```

```

const int CUT = 100, LEN = 4; // CUT >= log2(mod); A = [2 for
1..LEN] > CUT
int n; int arr[100020];
int solve(int idx){
    int val = 1; for (int i = n; i >= idx; i--){
        val = fpow(arr[i], val, (CUT+1)*(CUT+2), CUT+1);
    } return val;
}
int solve(int idx, int mod){
    if (mod == 1){ return 0; }
    if (idx == n){ return arr[idx] % mod; }
    if (n-idx+1 <= LEN){
        if (solve(idx+1) <= CUT){ return fpow(arr[idx],
            solve(idx+1), mod); }
    }
    int m = phi[mod];
    return fpow(arr[idx], solve(idx+1, m) + CUT*m, mod);
}

```

#### 4.9 • Iterating Floor of k/n

```

vector<pl2> v; // saves {floor(k/n), # of k}.
ll sq = (ll)sqrt(n);
for (ll i = 1; i <= sq; i++){
    ll val = n/i; if (v.empty() || v.back().fr != val){
        v.push_back({val, 1}); }
    else{ v.back().sc += 1; }
}
for (ll x = sq; x >= 1; x--){
    ll st = max(sq+1, n/(x+1)+1), ed = n/x;
    if (st <= ed){ v.push_back({x, ed-st+1}); }
}

```

#### 4.10 • Modular Inverses of 1 to N

```

inv[0] = 0; inv[1] = 1; for (int x = 2; x < X; x++){
    inv[x] = inv[mod%x]*(mod-mod/x) % mod;
}

```

## 5 Mathematics - Combinatorics & Probability Theory

### 5.1 • Basic Combinatorics

Let  $x^{(k)}$  be the product of  $x$  to  $x+k-1$ , and  $x_{(k)}$  be the product of  $x$  to  $x-k+1$ . Let  $C_{n,r} = \binom{n}{r}$  and  $H_{n,r} = \binom{n+r-1}{r}$ . Let  $P_{n,r} = n!/r!$ . Derangement is the number of permutation with  $\pi_i \neq i$  for all  $i$ .  $D_0 = 1$ ;  $D_1 = 0$ ;  $D_n = (n-1)(D_{n-1} + D_{n-2})$ .

Catalan Number is the number of valid Regular Bracket Sequence with  $n$  pairs.

$C_0 = 1$ ;  $C_{n+1} = (4n+2)C_n/(n+2)$ . Def:  $C_{n+1} = \sum_{k=0}^n C_k C_{n-k}$ .

Stirling Number of the First Kind is the number of permutation with  $n$  elements and  $k$  cycles.

$s_{0,0} = 1$ ;  $s_{n,0} = s_{0,k} = 0$ ;  $s_{n,k} = (n-1)s_{n-1,k} + s_{n-1,k-1}$ .

Stirling Number of the Second Kind is the number of ways to partition a set of  $n$  distinct objects onto  $k$  non-empty subsets.

$S_{0,0} = 1$ ;  $S_{n,0} = S_{0,k} = 0$ ;  $S_{n,k} = kS_{n-1,k} + S_{n-1,k-1}$ .

Bell Number is the number of ways to partition a set of  $n$  distinct objects.

$$B_n = \sum_{k=0}^n S_{n,k} = \sum_{k=0}^n C_{n,k} B_k.$$

Partition Number is the number of ways to partition  $n$  nondistinct objects into collection of  $k$  non-empty bunches.

$$p_{n,k} = p_{n-1,k-1} + p_{n-k,k} \cdot p_n = \sum_{k=1}^n p_{n,k}.$$

- Lucas's Theorem:  $C_{n,r} \equiv C_{n \bmod p, r \bmod p} C_{n/p, r/p} \pmod{p}$ .
- Vamdermonde's Identity:  $\sum_{k=0}^r C_{n,k} C_{m,r-k} = C_{n+m,r}$ .
- Hockey-Stick Identity: Assuming  $n > r$ ,  $\sum_{k=r}^n C_{k,r} = C_{n+1,r+1}$ .
- $\sum_{k=r}^n C_{n,k} C_{k,r} = 2^{n-k}$ .
- $\sum_{k=1}^n C_{n,k} C_{c-1,k-1} = C_{2n-1,n-1}$ .
- $\sum_{k \leq n} C_{n,k} a^{n-k} b^k = ((a+b)^n - (a-b)^n)/2$ .
- $\sum_{k=0}^n C_{r,k}/C_{n,k} = C_{n+1,n-r+1}/C_{n,r}$ .
- Number of involutions:  $I_0 = I_1 = 1$ ;  $I_n = I_{n-1} + (n-1)I_{n-2}$ .
- Number of permutations of size  $n$  where every cycles have length  $\leq k$ :  $T_{n,k} = n!$  if  $n \leq k$  else  $nT_{n-1,k} - (n-1)_{(k)}T_{n-k-1,k}$ .
- Catalan Number = Number of permutations of length  $n$  that avoids the pattern 123.
- Number of prefix RBS of length  $n+k$  with  $n$  opens and  $k$  closes:  $C_{n+k,k}(n-k+1)/(n+1)$ .
- Number of RBS of length  $2n$  with exactly  $k$  '('s:  $C_{n,k}C_{n,k-1}/n$ .
- Stirling Number of the Second Kind but each subset's size  $\geq r$ :  $S_{r,n+1,k} = kS_{r,n,k} + C_{n,r-1}S_{r,n-r+1,k-1}$ .
- Stirling Number of the Second Kind but pairwise distance  $\geq d$ :  $S_{n-d+1,k-d+1}$ , assuming  $n \geq k \geq d$ .
- Burnside: Let  $G$  be a group.  $r = \sum_{g \in G} |X_g|/|G|$ .  $|X_g|$  is the number of elements that  $gx = x$  holds.
- Linearity of Expectation:  $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ . If independent,  $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ .
- Variance:  $\text{Var}(X+a) = \text{Var}(X)$ ;  $\text{Var}(aX) = a^2\text{Var}(X)$ .
- Covariance:  $\text{cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ .
- $\text{Var}(X) = \text{cov}(X,X)$ ;  $\text{cov}(aX + bY, cW + dV) = a\text{cov}(X,W) + b\text{cov}(Y,W) + c\text{cov}(X,V) + d\text{cov}(Y,V)$ .
- $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2abc\text{cov}(X,Y)$ .

### Thirtyfold Way

Given  $n$  items, divide it up to  $k$  batches.  $n, k \geq 1$ . Batches can be empty on Problem 0, 1, 2.

- A: Distinguishable items onto Sequence of Lists.
- B: Distinguishable items onto Sequence of Sets.
- C: Indistinguishable items onto Sequence of Bunches.
- D: Distinguishable items onto Collection of Lists.
- E: Distinguishable items onto Collection of Sets.
- F: Indistinguishable items onto Collection of Bunches.
- 0: You are given sequence  $a$  that  $\sum ia_i = n$  and  $\sum a_i = k$ . There must be exactly  $a_i$  batches that have size  $i$ .
- 1: Each Batch must contain at most 1 item.
- 2: It's fine as long as you have  $k$  batches.
- 3: Each Batch must contain at least 1 item.

- 4: You are only given  $n$ , and are to find sum of Problem 3 for all  $b$ .

For Problem 0, define  $\alpha = \langle a_0, a_1, \dots, a_n \rangle$ . Let  $\iota = \langle 1, \dots, 1, 2, \dots, 2, \dots, n, \dots, n \rangle$  where  $x$  appears  $a_x$  times.

Let  $\alpha! = a_0!a_1!\dots a_n!$  and  $\alpha^{+!} = a_1!a_2!\dots a_n!$ . Let  $\iota! = 1!\dots 1!2!\dots 2!\dots n!\dots n!$ .

Let  $C(k, \alpha) = k!/\alpha!$  and  $C(n, \iota) = n!/\iota!$ .

|                                      | 0                                        | 1                                               |
|--------------------------------------|------------------------------------------|-------------------------------------------------|
| 2                                    | 3                                        | 4                                               |
| A<br>$k^{(n)}$                       | $n!C_{k,\alpha}$<br>$C_{n-1,k-1}n!$      | $k^{(n)}$<br>$2^{n-1}n!$                        |
| B<br>$k^n$                           | $C_{k,\alpha}C_{n,\iota}$<br>$k!S_{n,k}$ | $k^{(n)}$<br>$\sum_{k=1}^n k!S_{n,k}$           |
| C<br>$H_{k,n}$                       | $C_{k,\alpha}$<br>$C_{n-1,k-1}$          | $C_{k,n}$<br>$2^{n-1}$                          |
| D<br>$\sum_{i=1}^k C_{n-1,k-1}n!/i!$ | $m!/\alpha^{+!}$<br>$C_{n-1,k-1}n!/k!$   | $[n \leq k]$<br>$\sum_{k=1}^n C_{n-1,k-1}n!/k!$ |
| E<br>$\sum_{i=1}^k S_{n,i}$          | $C_{m,\iota}/\alpha^{+!}$<br>$S_{n,k}$   | $[n \leq k]$<br>$B_n$                           |
| F<br>$\sum_{i=1}^k p_{n,i}$          | 1<br>$p_{n,k}$                           | $[n \leq k]$<br>$p_n$                           |

### 5.2 • Sum of Binomial Coefficient

Let  $f(n, l, r) = \sum_{k=l}^r C_{n,k}$ . Change from  $f(n, l, r)$  to  $f(n', l', r')$  takes  $O(|n-n'| + |l-l'| + |r-r'|)$ .

- $f(n, l, r) \rightarrow f(n, l, r+1)$ : Add  $C_{n,r+1}$ .
- $f(n, l, r) \rightarrow f(n, l+1, r)$ : Subtract  $C_{n,l}$ .
- $f(n, l, r) \rightarrow f(n+1, l, r)$ : Multiply it by 2, add  $C_{n,l-1} - C_{n,r}$ .

### 5.3 • Lindström-Gessel-Viennot Lemma

Let  $w(P)$  be the product of the weights of the edges of the path  $P$ . Let  $D$  be the matrix with  $D_{i,j}$  be the sum of  $w(P)$ s with  $i \rightarrow j$ .

Then,  $\det(D) = \sum_{(P_1, \dots, P_N): S \rightarrow T} \text{sign}(\sigma(P)) \prod_{i=1}^N w(P_i)$ , where  $(P_1, \dots, P_N)$  is the  $N$  non-intersecting paths, with  $P_i = A_i \rightarrow B_{\sigma(i)}$ .

When every edge have weight 1,  $w(P) = 1$ . Therefore, the lemma above will just find the number of possible non-intersecting path (with parity of the permutation).

### 5.4 • Generating Function

Let  $A(x)$  be the GF of the sequence  $a_i$ .  $a$  is 0-indexed.

- If  $A$  is OGF,  $a_n = A^{(n)}(0)/n!$ .
- If  $A$  is EGF,  $a_n = A^{(n)}(0)$ .
- If  $C(x) = kA(x) + lB(x)$ , then  $c_i = ka_i + lb_i$ .
- If  $B(x) = xA'(x)$ , then  $b_i = ia_i$ .
- Let  $\langle b_0, b_1, \dots, b_{k-1}, b_k, b_{k+1}, \dots \rangle = \langle 0, 0, \dots, 0, a_0, a_1, \dots \rangle$ .
  - If  $A$  and  $B$  are both OGF, then  $B(x) = x^k A(x)$ .
  - If  $A$  and  $B$  are both EGF, then  $B(x) = \int^{(k)} A(x) d^k x$ .
- Let  $\langle b_0, b_1, \dots \rangle = \langle a_k, a_{k+1}, \dots \rangle$ .
  - If  $A$  and  $B$  are both OGF, then  $B(x) = (A(x) - a_0 - a_1x - \dots - a_{k-1}x^{k-1})/x^k$ .



- If  $A$  and  $B$  are both EGF, then  $B(x) = \frac{d^k}{dx^k} A(x)$ .
- Let  $C(x) = A(x)B(x)$ .
  - If  $A$  and  $B$  are both OGF, then  $c_i = \sum_{k=0}^i a_k b_{i-k} = (a*b)_i$ .
  - If  $A$  and  $B$  are both EGF, then  $c_i = \sum_{k=0}^i \binom{i}{k} a_k b_{i-k}$ .
- Let  $k$  be a positive integer, and let  $B(x) = A(x)^k$ .
  - If  $A$  and  $B$  are both OGF, then  $b_n = \sum_{i_1+i_2+\dots+i_k=n} a_{i_1} a_{i_2} \dots a_{i_k}$ .
  - If  $A$  and  $B$  are both EGF, then  $b_n = \sum_{i_1+i_2+\dots+i_k=n} \frac{n!}{i_1! i_2! \dots i_k!} a_{i_1} a_{i_2} \dots a_{i_k}$ .
- If  $A$  and  $B$  are both OGF with  $B(x) = \frac{A(x)}{1-x}$ , then  $b_i = a_0 + a_1 + \dots + a_i$ .
- $(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n$ .
- $\frac{1}{(1-x)^k} = \sum_{n=0}^{\infty} \binom{n+k-1}{n} x^n = \sum_{n=0}^{\infty} k H_n x^n$ .

## 5.5 • Number of Spanning Trees

Let  $c_{i,j}$  be the number of edges. Let  $M_{i,j}$  be  $-c_{i,j}$  if  $i \neq j$ , and  $\deg i$  otherwise. Then the number of spanning trees of the graph is the any cofactor of  $M$ .

For directed spanning tree,  $c_{i,j} = (i \rightarrow j)$  and  $c_{i,i} = \text{indeg } i$ . The number of spanning trees are the cofactor of  $(r, r)$ .

## 6 Mathematics - Algebra

### 6.1 • Basic Math

- $\sum_{n=0}^{\infty} n/2^n = 2$ .
- Lagrange's Identity:  $(\sum_i a_i^2)(\sum_i b_i^2) - (\sum_i a_i b_i)^2 = \sum_{i < j} (a_i b_j - a_j b_i)^2$ .
- Vieta's Formula: For a polynomial  $\sum_i a_i x^i$ ,  $\sum_{i_1 < i_2 < \dots < i_k} \prod_{j=1}^k r_{i_j} = (-1)^k a_{n-k}/a_n$ .
- Let Fibonacci Number be  $F_n = n$  if  $n \leq 1$  else  $F_{n-1} + F_{n-2}$ .
- $\sum_{k=0}^n F_k = F_{n+2} - 1$  and  $\sum_{k=0}^n F_k^2 = F_n F_{n+1}$ .
- $\sum_{k=0}^{n-1} F_{2k+1} = F_{2n}$  and  $\sum_{k=0}^n F_{2k} = F_{2n+1} - 1$ .
- $F_m F_{n+1} + F_{m-1} F_n = F_{m+n}$  and  $F_m F_{n+1} - F_{m-1} F_n = (-1)^n F_{m-n}$ .
- $F_{2n} = F_{n+1}^2 - F_{n-1}^2 = F_n(F_{n+1} + F_{n-1})$ .
- $\gcd(F_m, F_n) = F_{\gcd(m,n)}$ .
- Fibonacci Number mod  $p$  have period with at most  $6n$ .
- Given  $m > n > 0$  with  $\gcd(m, n) = 1$  and  $2 \mid mn$ ,  $(m^2 - n^2, 2mn, m^2 + n^2)$  is primitive Pythagorean triple.
- The number of Pythagorean triple with  $a, b \leq c = n$  is  $(\prod_{p \mid n} (2\alpha + 1) - 1)/2$ , where  $\alpha$  is the highest exponent s.t.  $p^\alpha$  divides  $n$ .
- Bernoulli Number  $b_n$ :  $\sum_{k=1}^n k^c = \frac{(-1)^k}{c+1} C_{c+1,k} b_k n^{c+1-k}$ ;  $b_0 = 1$ ;  $b_n = \frac{-1}{c+1} \sum_{r=0}^{c-1} C_{c+1,r} b_r$ .
- Vieta's Formula: Let  $r_1, r_2, \dots, r_n$  be the roots to  $P = \sum_{k=0}^n a_k x^k$ . Then  $\sum_{i_1 < i_2 < \dots < i_k} r_{i_1} r_{i_2} \dots r_{i_k} = (-1)^k a_{n-k}/a_n$ .
- Chicken McNugget Theorem: There are exactly  $(m-1)(n-1)/2$  different unexpressible numbers. Max is  $mn - m - n$ . Exactly one of  $k$  or  $mn - m - n - k$  is unexpressible.

- $1/1 - 1/2 + 1/3 - 1/4 + \dots + 1/(2n-1) - 1/(2n) = 1/(n+1) + 1/(n+2) + \dots + 1/(2n)$ .
- Lagrange Polynomial:  $\sum_{i=1}^n y_i \prod_{j=1, j \neq i}^n (x - x_j)/(x_i - x_j)$ .

### Linear Algebra

- Matrix:  $\text{tr}(AB) = \text{tr}(BA)$ ;  $(A^{-1})^T = (A^T)^{-1}$ .
- Determinant: Row Swap  $\times -1$ ; Multiply Row  $\times a$ ; Add to Other Row  $\pm 0$ .
- Determinant of a Diagonal Matrix is the product of the diagonal entries.
- $\det M = \det M^T$ ;  $\det(AB) = \det A \det B$ .
- Adjoint: Let  $(\text{adj} M)_{i,j}$  be the cofactor of  $M$  at  $(i, j)$ .  $M^{-1} = \text{adj} M / \det M$ .
- Characteristic Polynomial: Let  $P_M(\lambda) = \det(\lambda I - M)$ .
- Eigenvalue:  $\lambda$ s where  $P_M(\lambda) = 0$ ; Eigenvector:  $(M - \lambda I)X = 0$  for each  $\lambda$  we got.
- Least Square:  $A$  be a matrix and  $b$  be a vector.  $\min_{x \in \mathbb{R}^n} \|Ax - b\|^2$ ;  $A^T A x = A^T b$ .
- $AB = C$  using Randomization in  $O(N^2)$ : Create random vector  $v$ , and see  $A(Bv) = Cv$ .

### Calculus

- $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$ ;  $(f(x)/g(x))' = (f'(x)g(x) - f(x)g'(x))/g(x)^2$ .
- $(f(g(x)))' = f'(g(x))g'(x)$ .  $\int u dv = uv - \int v du$ .
- Let  $x = g(t)$ . Then  $\int f(x) dx = \int f(g(t))g'(t) dt$ .
- $(a^x)' = a^x \ln a$ ;  $\int a^x dx = a^x / \ln a + C$ .
- $(\log_b |x|)' = 1/(x \ln b)$ .  $\int 1/x dx = \ln |x| + C$ .
- $\sec, \csc, \cot$  is  $1/\cos, 1/\sin, 1/\tan$  resp.
- $\arcsin, \arccos, \arctan$  is inverse of  $\sin, \cos, \tan$  resp.
- $\sinh, \cosh, \tanh$  is  $(e^x - e^{-x})/2, (e^x + e^{-x})/2, \sinh/\cosh$  resp.
- $x$  axis Rotation Volume: Assume  $f(x) \geq 0$ ,  $f$  is continuous in  $[a, b]$ .  $\int_a^b 2\pi x f(x) dx$ .

| Derivative             | $f(x)$        | Integral (w/ $+C$ )              |
|------------------------|---------------|----------------------------------|
| $\cos x$               | $\sin x$      | $-\cos x$                        |
| $-\sin x$              | $\cos x$      | $\sin x$                         |
| $\sec^2 x$             | $\tan x$      | $\ln  \sec x $                   |
| $\sec x \tan x$        | $\sec x$      | $\ln  \sec x + \tan x $          |
| $-\csc x \cot x$       | $\csc x$      | $\ln  \csc x - \cot x $          |
| $-\csc^2 x$            | $\cot x$      | $\ln  \sin x $                   |
| $1/\sqrt{1-x^2}$       | $\arcsin x$   | $x \arcsin x + \sqrt{1-x^2}$     |
| $-1/\sqrt{1-x^2}$      | $\arccos x$   | $x \arccos x - \sqrt{1-x^2}$     |
| $1/(1+x^2)$            | $\arctan x$   | $x \arctan x - \ln(x^2+1)/2$     |
| $1/( x \sqrt{x^2-1})$  | $\sec^{-1} x$ | $x \sec^{-1} x - \cosh^{-1}  x $ |
| $-1/( x \sqrt{x^2-1})$ | $\csc^{-1} x$ | $x \csc^{-1} x + \cosh^{-1}  x $ |
| $-1/(1+x^2)$           | $\cot^{-1} x$ | $x \cos^{-1} x + \ln(x^2+1)/2$   |

### 6.2 • Sum of k-th Power of Integers

First, compute  $0^k$  to  $(k+1)^k$  with binary lifting in  $O(k \log k)$  total time.

If  $n \leq k+1$  then the answer is obvious. Otherwise, find the polynomial interpolation, and plug  $x = n$ .

### 6.3 • 2-Satisfiability

- OR:  $(a \vee b) \rightarrow (\neg a \implies b) \wedge (\neg b \implies a)$ .
- AND:  $(a \wedge b) \rightarrow (a \vee a) \wedge (b \vee b)$ .
- NAND:  $(a \overline{\wedge} b) \rightarrow \neg(a \wedge b) \rightarrow (\neg a \vee \neg b)$ .
- NOR:  $(a \overline{\vee} b) \rightarrow (\neg a \wedge \neg b)$ .
- XOR:  $(a \oplus b) \rightarrow (a \vee b) \wedge (\neg a \vee \neg b)$ .
- XNOR:  $(a \oplus b) \rightarrow (a \vee \neg b) \wedge (\neg a \vee b)$ .

### Picking at most 1 variable

3 variables  $(a, b, c)$  can be done with  $(\neg a \vee \neg b) \wedge (\neg b \vee \neg c) \wedge (\neg c \vee \neg a)$ . This can be expanded to make  $O(N^2)$  clauses.

$N$  variables  $(x_1, x_2, \dots, x_N)$  can be done with the following method. Let  $y_i$  be true if one of  $x_1, x_2, \dots, x_i$  is true. This can be modeled as following:

- $x_i \implies y_i$ .
- $y_i \implies y_{i+1}$ .
- $y_i \implies \neg x_{i+1}$ .

### Constructing a Solution

```

for (int i = -n; i <= +n; i++){
    if (i == 0){ continue; }
    if (ord[i+Z] == 0){ dfs(i+Z); }
}
for (int i = 1; i <= n; i++){
    if (scc[i+Z] == scc[-i+Z]){ cout << -1; return; }
}
for (int i = -n; i <= +n; i++){
    if (i == 0){ continue; }
    int v = i+Z; for (int w : adj[v]){
        int pv = scc[v], pw = scc[w]; if (pv == pw){ continue; }
        dag[pv].push_back(pw); ind[pw] += 1;
    }
}
queue<int> q; for (int i = -n; i <= +n; i++){
    if (i == 0){ continue; }
    int v = i+Z; if (scc[v] != v){ continue; }
    if (ind[v] == 0){ q.push(v); }
} while (!q.empty()){
    int v = q.front(); q.pop();
    for (int p : sccv[v]){
        if (ans[p] != 0){ continue; }
        ans[p] = -1; ans[-(p-Z)+Z] = +1;
    }
    for (int w : dag[v]){
        ind[w] -= 1; if (ind[w] == 0){ q.push(w); }
    }
}

```

## 6.4 • Minimizing Quadratic Pseudo-Boolean Function

Given  $N$  boolean variables  $x_1, x_2, \dots, x_N$ , minimize the cost  $f(x_1, x_2, \dots, x_N) = c + \sum_i b_i(x_i) + \sum_{i < j} a_{i,j}(x_i, x_j)$ . Function  $a$  must satisfy the condition  $a(0,0) + a(1,1) \leq a(0,1) + a(1,0)$ .

This can be modeled as following. Let there be source  $s$ , sink  $t$ , and  $N$  vertices  $x_1, x_2, \dots, x_N$ .

- $c$  is trivial.
- $b_i(x_i = 0)$  can be modeled as  $x_i \rightarrow t$  with weight  $b_i(0)$ .
- $b_i(x_i = 1)$  can be modeled as  $s \rightarrow x_i$  with weight  $b_i(1)$ .
- $a_{i,j}(x_i, x_j)$  can be modeled as following:
  - $s \rightarrow v_i$  with weight  $a_{i,j}(1,0)$ .
  - $v_i \rightarrow t$  with weight  $a_{i,j}(0,0)$ .
  - $s \rightarrow v_j$  with weight  $a_{i,j}(1,1) - a_{i,j}(1,0)$ .
  - $v_i \rightarrow v_j$  with weight  $a_{i,j}(0,1) + a_{i,j}(1,0) - a_{i,j}(0,0) - a_{i,j}(1,1)$ .

Find the minimum cut, then assign  $x_i = 0$  if it's in  $s$ -side, and assign  $x_i = 1$  if it's in  $t$ -side.

```
inline void psh(int v, bool b, ll x){
    if (b == true){ adj[src][v] += x; }
    if (b == false){ adj[v][snk] += x; }
}
inline void psh(int v, ll w0, ll w1){
    psh(v, false, w0); psh(v, true, w1);
}
inline void psh(int i, int j, ll w00, ll w01, ll w10, ll w11){
    psh(i, true, w10); psh(i, false, w00);
    psh(j, true, w11-w10);
    adj[i][j] += w01+w10-w00-w11;
}

ll ans = 0; for (int i = 1; i <= n; i++){
    ll &w0 = adj[i][snk], &w1 = adj[src][i];
    ll mn = min(w0, w1); ans += mn; w0 -= mn; w1 -= mn;
}
```

## 6.5 • Fast Fourier Transform

```
void dft(vector<cpl>& arr, bool inv = false){ int n = arr.size();
    for (int j=0, i=1; i < n; i++){ int bit = n>>1;
        while (j & bit){ j ^= bit; bit >>= 1; } j ^= bit;
        if (i < j){ swap(arr[i], arr[j]); }
    }
    for (int l = 1; l < n; l <= l*2){
        if (complex){
            ld ang = PI / l; if (inv){ ang *= -1; }
            cpl w = {cos(ang), sin(ang)};
        }
        else{
            ll w = fpow(r, (mod-1)/(2*l)); if (inv){ w = finv(w); }
            // r is primitive root of mod. 998244353 -> 3
        }
    }
}
```

```
for (int i = 0; i < n; i += l<<1){
    cpl wp = 1; for (int j = 0; j < l; j++){
        cpl a = arr[i+j], b = arr[i+j + l] * wp;
        arr[i+j] = a+b; arr[i+j + l] = a-b; wp *= w;
    }
}
if (inv){ for (int i = 0; i < n; i++){ arr[i] /= n; } }
}
void mul(vector<cpl>& arr, vector<cpl>& brr){
    int n = max(arr.size(), brr.size()); n = bitp(n) * 2; // power of 2
    arr.resize(n); brr.resize(n);
    dft(arr); dft(brr);
    for (int i = 0; i < n; i++){ arr[i] *= brr[i]; }
    dft(arr, -1);
}
```

## Fast Welsh-hadamard Transform

```
void fwht(vector<ll>& arr, bool inv = false){
    int n = arr.size();
    for (int l = 1; l < n; l*=2){
        for (int p = 0; p < n; p+=l*2){
            for (int i = 0; i < l; i++){
                ll a = arr[p+i], b = arr[p+l+i];
                if (XOR){ arr[p+i] = (a+b)%mod; arr[p+l+i] = (a-b+mod)%mod; }
                if (AND){ arr[p+i] = (a + (inv ? -b : b) + mod) % mod; }
                if (OR){ arr[p+l+i] = (b + (inv ? -a : a) + mod) % mod; }
            }
        }
        if (inv && XOR){ for (int i = 0; i < n; i++){ arr[i] *= invn; } }
    }
}
```

## 6.6 • Floor Sum of Arithmetic Progression

Let  $f(a, b, c, n) = \sum_{x=0}^n \left\lfloor \frac{ax+b}{c} \right\rfloor$ .

If  $a \geq c$  or  $b \geq c$ , then  $f(a, b, c, n) = \frac{n(n+1)}{2} \left\lfloor \frac{a}{c} \right\rfloor + (n+1) \left\lfloor \frac{b}{c} \right\rfloor + f(a \bmod c, b \bmod c, c, n)$ .

Otherwise, we have  $f(a, b, c, n) = nm - f(c, c-b-1, a, m-1)$  where  $m = \left\lfloor \frac{an+b}{c} \right\rfloor$ .

## 6.7 • Linear Programming

If possible, minimize the number of variables.

Primal: Maximize  $\mathbf{c}^T \mathbf{x}$  subject to  $A\mathbf{x} \leq \mathbf{b}$  and  $\mathbf{x} \geq 0$ .

Dual: Maximize  $-\mathbf{b}^T \mathbf{y}$  subject to  $-\mathbf{A}^T \mathbf{y} \leq -\mathbf{c}$  and  $\mathbf{y} \geq 0$ .

- **Time Complexity:**  $O(NM)$  per pivot,  $O(2^N)$  at worst.  $O(N^3)?$ .

```
int n, m; vector<int> slk, piv;
matrix<ld> mat;
void init(const matrix<ld> &a, const vector<ld> &b, const vector<ld> &c){
```

```
n = c.size(); m = b.size();
slk = vector<int>(n+1); piv = vector<int>(m);
mat = matrix<ld>(m+2, vector<ld>(n+2));
for (int i = 0; i < m; i++){
    for (int j = 0; j < n; j++){ mat[i][j] = a[i][j]; }
}
for (int i = 0; i < m; i++){
    piv[i] = n+i; mat[i][n] = -1; mat[i][n+1] = b[i];
}
for (int j = 0; j < n; j++){
    slk[j] = j; mat[m][j] = -c[j];
} slk[n] = -1; mat[m+1][n] = 1;
}
void pivot(int r, int s){
    ld *a = mat[r].data(); ld inva = 1/a[s];
    for (int i = 0; i < m+2; i++){
        if (i == r || abs(mat[i][s]) <= eps){ continue; }
        ld *b = mat[i].data(); ld invb = b[s]*inva;
        for (int j = 0; j < n+2; j++){ b[j] -= a[j]*invb; }
        b[s] = a[s]*invb;
    }
    for (int j = 0; j < n+2; j++){ if (j != s){ mat[r][j] *= inva; } }
    for (int i = 0; i < m+2; i++){ if (i != r){ mat[i][s] *= -inva; } }
    mat[r][s] = inva; swap(piv[r], slk[s]);
}
bool simplex(int typ){
    int x = m+typ-1; while (1){
        int s = -1, r = -1; for (int j = 0; j < n+1; j++){
            if (slk[j] == -typ){ continue; }
            if (s == -1 || mkp(mat[x][j], slk[j]) < mkp(mat[x][s], slk[s])){ s = j; }
        } if (mat[x][s] >= -eps){ return 1; }
        for (int i = 0; i < m; i++){
            if (mat[i][s] <= eps){ continue; }
            if (r == -1 || mkp(mat[i][n+1]/mat[i][s], piv[i]) < mkp(mat[r][n+1]/mat[r][s], piv[r])){ r = i; }
        } if (r == -1){ return 0; }
        pivot(r, s);
    }
}
pair<ld, vector<ld>> solve(){
    int r = 0; for (int i = 1; i < m; i++){
        if (mat[i][n+1] < mat[r][n+1]){ r = i; }
    }
    if (mat[r][n+1] < -eps){
        pivot(r, n); if (!simplex(2)){ return {-inf, {}}; }
        if (mat[m+1][n+1] < -eps){ return {-inf, {}}; }
        for (int i = 0; i < m; i++){
            if (piv[i] != -1){ continue; }
            int s = 0; for (int j = 1; j < n+1; j++){
                if (s == -1 || mkp(mat[i][j], slk[j]) < mkp(mat[i][s], slk[s])){ s = j; }
            } pivot(i, s);
        }
    }
}
```

```
bool chk = simplex(1);
vector<ld> x(n); for (int i = 0; i < m; i++){
    if (piv[i] < n){ x[piv[i]] = mat[i][n+1]; }
    } return {chk ? mat[m][n+1] : inf, x};
}
```

## 6.8 • Multipoint Evaluation

```
vector<ll> seg[524288];
vector<ll> pos; void init(int ni, int ns, int ne){
    if (ns == ne){ seg[ni] = vector<ll>{(mod-pos[ns])%mod,
    1}); return; }
    int nm = ns+ne >> 1;
    init(ni<<1, ns, nm); init(ni<<1|1, nm+1, ne);
    seg[ni] = pmul(seg[ni<<1], seg[ni<<1|1]);
} inline void init(vector<ll> p){ pos = p; init(1, 0,
p.size()-1); }
vector<ll> res; void eval(int ni, int ns, int ne, vector<ll>
arr){
    if (ns == ne){ res[ns] = eval(arr, pos[ns]); return; }
    int nm = ns+ne >> 1;
    eval(ni<<1, ns, nm, pmod(arr, seg[ni<<1])); eval(ni<<1|1,
nm+1, ne, pmod(arr, seg[ni<<1|1]));
} inline vector<ll> eval(vector<ll> p){
    init(p); res.resize(p.size()); eval(1, 0, p.size()-1, arr);
}
```

## 7 Mathematics - Geometry

$y$  is first,  $x$  is second. Polygon is Counterclockwise.

### 7.1 • Basic Geometry

- Distance of Point  $(x', y')$  and Line  $ax + by + c = 0$ :  $\frac{|ax' + by' + c|}{\sqrt{a^2 + b^2}}$ .
- Volume of Polygonal Pyramid: Base  $\times$  Height/3.
- Volume of a Sphere:  $4\pi r^3/3$ . Surface Area:  $4\pi r^2$ .
- Pick's Theorem: if  $i$  inside and  $b$  boundary then area is  $i + b/2 - 1$ .
- $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$ ;  $\sin^2(\theta/2) = (1 - \cos \theta)/2$ .
- $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$ ;  $\cos^2(\theta/2) = (1 + \cos \theta)/2$ .
- $\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$ ;  $\tan^2(\theta/2) = \frac{1 - \cos \theta}{1 + \cos \theta}$ .
- Let  $a, b, c$  be the length;  $A, B, C$  be the angle of a triangle.
- Let  $s = (a + b + c)/2$ . Then area  $S = \sqrt{s(s-a)(s-b)(s-c)}$ .  
 $S = (bc \sin A)/2$ .
- Radius of Circumcircle  $R = abc/4S$ ; Radius of Incircle  $r = S/s$ .
- $a/\sin A = 2R$ ;  $a^2 = b^2 + c^2 - 2bc \cos A$ ;  $\frac{a+b}{a-b} = \frac{\tan((A+B)/2)}{\tan((A-B)/2)}$ .
- Let  $A = b^2 + c^2 - a^2$ .  $B$  and  $C$  is defined similarly. Map  $(a)$  as  $(a : b : c)$ , cyclic.
- Incenter:  $(a)$ ; Circumcenter:  $(a^2A)$ ; Centroid:  $(1)$ ; Orthocenter:  $BC$ .
- Excenter(A):  $(-a : b : c)$ ; Nine-point:  $(a^2(b^2 + c^2) - (b^2 - c^2)^2)$ .
- Given a center  $P$ , Let  $D$  be the intersection of  $AP$  and  $BC$ .  $E$  and  $F$  is defined similarly.
- Incenter  $I$ :  $AI$  is also an angle bisector. Circumcenter  $O$ : Circle passes  $A, B$ , and  $C$ .
- Orthocenter  $H$ :  $AH$  and  $BC$  are orthic i.e. angle is 90 deg. Centroid:  $AG : GD = 2 : 1$  and  $|BD| = |DC|$ .

- $\text{proj}_{\mathbf{u}} \mathbf{v} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$ .

### 7.2 • Nearest Pair of Point

```
sort(arr+1, arr+n+1, [](p12 p1, p12 p2){ return p12(p1.sc,
p1.fr) < p12(p2.sc, p2.fr); });
int ans = dis(arr[1], arr[2]);
set<p12> s; s.insert(arr[1]); s.insert(arr[2]);
int st = 1; for (int i = 3; i <= n; i++){
    while (st < i){
        ll x = (p[st].sc-p[i].sc); if (x*x <= ans){ break; }
        else{ s.erase(arr[st]); st += 1; }
    } int d = (int)sqrt(ans)+1;
    auto l = s.lower_bound({p[i].fr-d, -INF-1});
    auto r = s.upper_bound({p[i].fr+d, INF+1});
    for (auto j = l; j != r; j++){ ans = min(ans, dis(p[i],
*j)); }
    s.insert(p[i]);
}
```

### 7.3 • Line Intersection

```
inline ll cross(const p12& v1, const p12& v2){ return
v1.sc*v2.fr - v2.sc*v1.fr; }
inline p12 ltov(const p14& l){ return {l.sc.fr-l.fr.fr,
l.sc.sc-l.fr.sc}; }
int intersect(p14 l1, p14 l2){
    p12 p1 = l1.fr, p2 = l1.sc; p12 p3 = l2.fr, p4 = l2.sc;
    int c123 = ccw(p1, p2, p3), c124 = ccw(p1, p2, p4);
    int c341 = ccw(p3, p4, p1), c342 = ccw(p3, p4, p2);
    if (c123 == 0 && c124 == 0){
        if (p1 > p2){ swap(p1, p2); } if (p3 > p4){ swap(p3, p4); }
        if (p2 < p3 || p4 < p1){ /* No Intersection */ }
        if (p2 == p3 || p4 == p1){ /* Endpoint */ }
        /* Infinitely Many */
    }
    int c12 = c123*c124, c34 = c341*c342;
    if (c12 > 0 || c34 > 0){ /* No Intersection */ }
    if (c12 == 0 || c34 == 0){ /* Endpoint */ }
    /* Mid-line */
}
pd2 intersection(const p14& l1, const p14& l2){
    p12 p1 = l1.fr, p2 = l1.sc; p12 p3 = l2.fr, p4 = l2.sc;
    if (p1 == p3 || p1 == p4){ return p1; } if (p2 == p3 || p2
== p4){ return p2; }
    p12 v1 = ltov(l1), v2 = ltov(l2); p12 d = {p3.fr-p1.fr,
p3.sc-p1.sc};
    ld a = (ld)cross(d, v2) / cross(v1, v2);
    return {p1.fr + v1.fr*a, p1.sc + v1.sc*a};
}
```

### 7.4 • Convex Hull

```
for (int i = 1; i <= n; i++){
    if (arr[i] > arr[i-1]){ swap(arr[i], arr[i-1]); }
} p12 p0 = arr[1]; sort(arr+2, arr+n+1, [&p0](const p12& p1,
const p12& p2){
    int res = ccw(p0, p1, p2); if (res != 0){ return res > 0; }
}
```

```
return dis(p0, p1) < dis(p0, p2);
});
vector<p12> stk; for (int i = 1; i <= n; i++){
    p12 p3 = arr[i]; while (stk.size() >= 2){
        p12 p1 = *prev(prev(stk.end())), p2 = *prev(stk.end());
        if (ccw(p1, p2, p3) <= 0){ stk.pop_back(); } else{ break; }
    } // < 0 if linear
    } stk.push_back(p3);
}
for (int i = n-1; i > 1; i--){ // if linear only
    if (ccw(stk.front(), arr[i], stk.back()) == 0){
        stk.push_back(arr[i]); }
    else{ break; }
}
```

### 7.5 • Point in Polygon

#### Convex Polygon

```
int st = 2, ed = n; while (st+1 <= ed-1){
    int mid = st+ed >> 1;
    if (ccw(arr[1], arr[mid], p) == +1){ st = mid; } else{ ed =
mid; }
} p12 p1 = arr[1], p2 = arr[st], p3 = arr[st+1];
int r12 = ccw(p1, p2, p), r23 = ccw(p2, p3, p), r31 = ccw(p3,
p1, p);
if (r12 < 0 || r23 < 0 || r31 < 0){ /* Outside */ }
else{
    if (r23 == 0 || r12 == 0 && st == 2 || r31 == 0 && st ==
n-1){ /* Line */ }
    else{ /* Inside */ }
}
```

#### Non-convex Polygon

If the point is on the line, then whatever do what the problem says.  
Otherwise, send the straight line  $(x, y)$  to  $(x + X, y + X + 1)$  with  $X$  being sufficiently large integer. Then, count the number of intersections.

### 7.6 • Rotating Calipers

```
int j = 0; for (int i = 0; i < sp; i++){
    while (j < sp){
        int ip = (i+1) % sp, jp = (j+1) % sp;
        p12 l = { arr[ip].fr - arr[i].fr, arr[ip].sc - arr[i].sc
};
        p12 r = { arr[jp].fr - arr[j].fr, arr[jp].sc - arr[j].sc
};
        if (ccw({0, 0}, l, r) < 0){ break; }
        ans = max(ans, dist(i, j)); j += 1;
    } if (j < sp){ ans = max(ans, dist(i, j)); }
}
```

### 7.7 • Tangent Line of a Convex Polygon

Let  $i_1$  and  $i_2$  be the result of tangent(p, -1) and tangent(p, +1) respectively. The visible vertices from  $p$  is  $P_{i_1}$  to  $P_{i_2}$ . Note that the index is cyclic i.e. if  $i_1 > i_2$  then the visible vertices are  $P_{i_1}$  to  $P_n$  and  $P_1$  to  $P_{i_2}$ .

If there are multiple tangent line on the single direction, it returns the nearest one. Test prv and nxt resp. for farthest in dir  $-1$  and  $+1$ , resp.

```
int n; pl2 arr[100020];
inline int nxt(int i){ return i==n ? 1 : i+1; }
inline int prv(int i){ return i==1 ? n : i-1; }
inline bool locate(const pl2& p, const pl2& p1, const pl2& p2, const pl2& p3, int dir){
    return dir*ccw(p, p1, p2) <= 0 && dir*ccw(p, p2, p3) >= 0;
}
int fix(const pl2& p, int idx, int dir){
    if (dir > 0 && ccw(p, arr[idx], arr[prv(idx)]) == 0){
        pl2 p1 = arr[idx], p2 = arr[prv(idx)]; if (p1 > p2){
            swap(p1, p2); }
        if (p1 <= p && p <= p2){ return idx; } else{ return prv(idx); }
    }
    if (dir < 0 && ccw(p, arr[idx], arr[nxt(idx)]) == 0){
        pl2 p1 = arr[idx], p2 = arr[nxt(idx)]; if (p1 > p2){
            swap(p1, p2); }
        if (p1 <= p && p <= p2){ return idx; } else{ return nxt(idx); }
    }
    return idx;
}
int tangent(const pl2& p, int dir){
    if (locate(p, arr[prv(1)], arr[1], arr[nxt(1)], dir)){
        return fix(p, 1, dir); }
    int st = 1, ed = n+1; while (st+1 <= ed-1){
        int mid = st+ed >> 1;
        if (locate(p, arr[prv(mid)], arr[mid], arr[nxt(mid)], dir)){ return fix(p, mid, dir); }
        if (dir*ccw(p, arr[st], arr[nxt(st)]) < 0){
            if (dir*ccw(p, arr[mid], arr[nxt(mid)]) > 0){ ed = mid; }
            else if (dir*ccw(p, arr[mid], arr[st]) > 0){ st = mid; }
            else{ ed = mid; }
        } else{
            if (dir*ccw(p, arr[mid], arr[nxt(mid)]) < 0){ st = mid; }
            else if (dir*ccw(p, arr[mid], arr[st]) < 0){ st = mid; }
            else{ ed = mid; }
        }
    }
    if (st != 1 && locate(p, arr[prv(st)], arr[st], arr[nxt(st)], dir)){ return fix(p, st, dir); }
    if (ed != n+1 && locate(p, arr[prv(ed)], arr[ed], arr[nxt(ed)], dir)){ return fix(p, ed, dir); }
    return -1;
}
```

## 7.8 • Minkowski Sum of Convex Polygons

```
pl2 operator+(const pl2& p1, const pl2& p2){ return
{p1.y+p2.y, p1.x+p2.x}; }
```

```
pl2 operator-(const pl2& p1, const pl2& p2){ return
{p1.y-p2.y, p1.x-p2.x}; }
```

```
for (int t = 1; t <= n; t++){
    for (int i = 0; i < m; i++){
        int j = i+1; if (j >= m){ j -= m; }
        pl2 p1 = stk[i], p2 = stk[j];
        pl2 d = p2-p1; d = {-d.x, d.y};
        evt.push_back({d, p2-p1});
    }
    for (int i = 0; i < m; i++){
        int j1 = i-1; if (j1 < 0){ j1 += m; }
        int jr = i+1; if (jr >= m){ jr -= m; }
        pl2 p1 = stk[j1], p = stk[i], pr = stk[jr];
        pl2 dl = p-p1, dr = pr-p; dl = {-dl.x, dl.y}; dr = {-dr.x, dr.y};
        if (pl2(-dl.y, dl.x) > pl2(0, 0) && pl2(-dr.y, dr.x) < pl2(0, 0)){ res = res+p; }
    }
}
sort(evt.begin(), evt.end(), [](const pl4& pp1, const pl4& pp2){
    pl2 p1 = pp1.fr, p2 = pp2.fr;
    pl2 o = {0, 0}; if ((p1 > 0) != (p2 > 0)){ return (p1 > 0) > (p2 > 0); }
    int res = ccw(0, p1, p2); return res > 0;
});
ans = max(ans, dis(res)); for (pl4& p : evt){
    if (p.fr.y == 0 && p.fr.x > 0){ continue; }
    res = res+p.sc; ans = max(ans, dis(res));
} for (pl4& p : evt){
    if (!(p.fr.y == 0 && p.fr.x > 0)){ continue; }
    res = res+p.sc; ans = max(ans, dis(res));
} cout << ans;
```

## 7.9 • Minimum Enclosing Circle

```
inline pd2 circumcenter(const pd2& p1, const pd2& p2, const pd2& p3){
    ld d1 = dis(p2, p3), d2 = dis(p3, p1), d3 = dis(p1, p2);
    d1 *= d1; d2 *= d2; d3 *= d3;
    ld w1 = d1*(d2+d3-d1), w2 = d2*(d3+d1-d2), w3 = d3*(d1+d2-d3);
    ld w = w1+w2+w3;
    w1 /= w; w2 /= w; w3 /= w;
    return pd2(w1*p1.fr + w2*p2.fr + w3*p3.fr, w1*p1.sc + w2*p2.sc + w3*p3.sc);
}
```

```
shuffle(arr+1, arr+n+1, gen);
pd2 o = {0, 0}; ld r = 0; for (int i = 1; i <= n; i++){
    if (inside(o, r, arr[i])){ continue; }
    o = arr[i]; r = 0;
    for (int j = 1; j < i; j++){
        if (inside(o, r, arr[j])){ continue; }
        o = {(arr[i].fr+arr[j].fr)/2, (arr[i].sc+arr[j].sc)/2}; r = dis(o, arr[i]);
        for (int k = 1; k < j; k++){
            if (inside(o, r, arr[k])){ continue; }
```

```
o = circumcenter(arr[i], arr[j], arr[k]); r = dis(o, arr[i]);
    }
} // center 0, radius R.
```

## 7.10 • Bulldozer Trick

update(pi): updates pi-th element on the data structure.

```
sort(arr+1, arr+n+1); for (int i = 1; i <= n; i++){ pos[i] = i; } // min y first
for (int i = 1; i <= n; i++){
    for (int j = i+1; j <= n; j++){
        pl2 p = {arr[i].fr.fr-arr[j].fr.fr, arr[i].fr.sc-arr[j].fr.sc};
        p = {p.sc, -p.fr};
        if (p.sc < 0 || p.sc==0 && p.fr > 0){ p = {-p.fr, -p.sc}; }
        lin.push_back({p, {i, j}}); // p as directional vector, min y first
    }
}
sort(lin.begin(), lin.end(), [](pl2i2 p1, pl2i2 p2){
    int res = ccw(pl2(0, 0), p1.fr, p2.fr); // ccw, if same then index
    if (res != 0){ return res > 0; } else{ return p1.sc < p2.sc; }
});
int ans = solve(state); for (pl2i2 p : lin){
    int i = p.sc.fr, j = p.sc.sc;
    int pi = pos[i], pj = pos[j];
    swap(arr[pi], arr[pj]); swap(pos[i], pos[j]); update(pi); update(pj);
    ans = ans or newState;
}
```

## 7.11 • Halfplane Intersection

$\mathbf{v}$  contains  $(y_1, x_1), (y_2, x_2)$ . Left side of  $\mathbf{v} = p_2 - p_1$  is the halfplane. Counterclockwise convex polygon  $P = \langle p_1, p_2, \dots, p_n \rangle$  will have intersection equivalent to  $P$ .

```
const pl2 o = {0, 0};
const ld eps = 1e-9;
const ll INF = 1e9;
inline bool eqf(ld a, ld b){ return abs(a-b)/max<ld>({abs(a), abs(b), 1}) <= eps; }
inline int sgn(ll x){ return (x > 0) - (x < 0); }
inline int sgn(ld x){ return eqf(x, 0) ? 0 : (x > 0) - (x < 0); }

inline int ccw(const pl2& p1, const pl2& p2, const pl2& p3){
    ll a = p1.sc*p2.fr + p2.sc*p3.fr + p3.sc*p1.fr;
    ll b = p1.sc*p3.fr + p3.sc*p2.fr + p2.sc*p1.fr;
    return sgn(a-b);
}
```



```

inline int ccwd(const pd2& p1, const pd2& p2, const pd2& p3){
    ld a = p1.sc*p2.fr + p2.sc*p3.fr + p3.sc*p1.fr;
    ld b = p1.sc*p3.fr + p3.sc*p2.fr + p2.sc*p1.fr;
    return sgn(a-b);
}

inline ld area(const pd2& p1, const pd2& p2, const pd2& p3){
    ld a = p1.sc*p2.fr + p2.sc*p3.fr + p3.sc*p1.fr;
    ld b = p1.sc*p3.fr + p3.sc*p2.fr + p2.sc*p1.fr;
    return abs(a-b)/2;
}

inline ll cross(const pl2& v1, const pl2& v2){ return
v1.sc*v2.fr - v2.sc*v1.fr; }

inline pl2 ltov(const pl4& l){ return pl2{l.sc.fr-l.fr.fr,
l.sc.sc-l.fr.sc}; }

pair<bool, pd2> intersection(const pl4& l1, const pl4& l2){
    pl2 v1 = ltov(l1), v2 = ltov(l2);
    if (ccw(0, v1, v2) == 0){ return {0, {0, 0}}; }
    pl2 p1 = l1.fr, p2 = l2.fr; pl2 dp = {p2.fr-p1.fr,
p2.sc-p1.sc};
    ld a = (ld)cross(dp, v2) / cross(v1, v2);
    return {1, {p1.fr + v1.fr*a, p1.sc + v1.sc*a}};
}

bool out(const pd2& p, const pl4& l){ return ccwd(l.fr, l.sc,
p) <= 0; }

bool out(const pl4& l1, const pl4& l2, const pl4& l){
    auto res = intersection(l1, l2); if (!res.fr){ return 0; }
    return out(res.sc, l);
}

vector<pd2> halfplane(vector<pl4> v){
    {
        pl2 p1 = {-INF, -INF}, p2 = {-INF, INF};
        pl2 p3 = {INF, INF}, p4 = {INF, -INF};
        v.push_back({p1, p2}); v.push_back({p2, p3});
        v.push_back({p3, p4}); v.push_back({p4, p1});
    }
    sort(v.begin(), v.end(), [](const pl4& l1, const pl4& l2){
        pl2 v1 = ltov(l1), v2 = ltov(l2);
        if ((v1 > 0) != (v2 > 0)){ return (v1 > 0) > (v2 > 0); }
        return ccw(0, v1, v2) > 0;
    });
    deque<pl4> dq; for (const pl4& l : v){
        while (dq.size() >= 2){
            int dql = dq.size();
            if (out(dq[dql-2], dq[dql-1], l)){ dq.pop_back(); }
            else{ break; }
        }
        while (dq.size() >= 2){
            int dql = dq.size();
            if (out(dq[0], dq[1], l)){ dq.pop_front(); }
            else{ break; }
        }
        int dql = dq.size(); if (dql >= 1){
            pl2 v1 = ltov(l), v2 = ltov(dq[dql-1]);
            if (ccw(0, v1, v2) == 0){
                if ((v1 > 0) == (v2 > 0)){

```

```

                pl2 p1 = l.fr;
                if (!out(p1, dq[dql-1])){ dq.pop_back();
                dq.push_back(l); }
                continue;
            } else{ return {}; }
        }
        if (dql < 2 || !out(dq[dql-1], l, dq[0])){
            dq.push_back(l); }
        int dql = dq.size(); if (dql < 3){ return {}; }
        vector<pd2> ans; for (int i = 0; i < dql; i++){
            int j = i+1; if (j == dql){ j = 0; }
            pl4 l1 = dq[i], l2 = dq[j]; auto res = intersection(l1,
            l2);
            if (res.fr){ ans.push_back(res.sc); }
        } return ans;
    }
}

```

## 7.12 • Shamos-Hoeoy

Simple Polygon version. For Full non-intersection follow the comments instead.

```

inline ld f(const pl4& l, ll x){
    return (ld)(l.sc.fr-l.fr.fr)/(l.sc.sc-l.fr.sc) *
(x-l.fr.sc) + l.fr.fr;
}

ll ptr; bool cmp4(const pl4& l1, const pl4& l2){
    ld p1 = f(l1, ptr), p2 = f(l2, ptr);
    return p1 < p2;
} bool cmp(const pl5& p1, const pl5& p2){ return cmp4(p1.fr,
p2.fr); }

// usual No Intersection / Yes Intersection if fully
nonintersected.
int intersect(const pl4& l1, const pl4& l2){
    pl2 p1 = l1.fr, p2 = l1.sc, p3 = l2.fr, p4 = l2.sc;
    int c123 = ccw(p1, p2, p3), c124 = ccw(p1, p2, p4);
    int c341 = ccw(p3, p4, p1), c342 = ccw(p3, p4, p2);
    if (c123 == 0 && c124 == 0){
        if (p1 > p2){ swap(p1, p2); } if (p3 > p4){ swap(p3, p4); }
    }
    if (p2 <= p3 || p4 <= p1){ return 0; } else{ return 1; }
}
if (p1 == p3 || p2 == p4){ return p1 == p3 ? -1 : -2; }
return c123 != c124 && c341 != c342;
} int intersect(const pl5& l1, const pl5& l2){ return
intersect(l1.fr, l2.fr); }

pl4 prr[100020]; pl4 pos[200020];
multiset<pl5, decltype(&cmp)> lin(cmp);
pi2 solve(int n){
    sort(pos+1, pos+n+1); lin.clear();
    for (int i = 1; i <= n+n; i++){
        ll x = pos[i].fr.fr; bool ed = pos[i].fr.sc;
        ll y = pos[i].sc.fr; int idx = pos[i].sc.sc;
        if (prr[idx].fr.sc == prr[idx].sc.sc){ continue; }
        pl5 l = {prr[idx], idx}; ptr = x;
        if (!ed){

```

```

// auto it = lin.insert(l);
// if (it != lin.begin()){ if (intersect(*it,
*prev(it)){ return {it->sc, prev(it)->sc}; } }
// if (next(it) != lin.end()){ if (intersect(*it,
*next(it)){ return {it->sc, next(it)->sc}; } }
bool flg = 0; auto it = lin.insert(l); again:
if (it != lin.begin()){
    int res = intersect(*it, *prev(it));
    if (res == 1){ return {it->sc, prev(it)->sc}; }
    if (res == -1){
        lin.erase(it); ptr += 1; it = lin.insert(l); ptr -=
        1;
        if (!flg){ flg = 1; goto again; }
    }
}
if (next(it) != lin.end()){
    int res = intersect(*it, *next(it));
    if (res == 1){ return {it->sc, next(it)->sc}; }
    if (res == -1){
        lin.erase(it); ptr += 1; it = lin.insert(l); ptr -=
        1;
        if (!flg){ flg = 1; goto again; }
    }
} else{
    // auto it = lin.lower_bound(l);
    // if (it != lin.begin() && next(it) != lin.end()){
    //     if (intersect(*prev(it), *next(it))){ return
    {prev(it)->sc, next(it)->sc}; }
    // } lin.erase(it);
    auto it = lin.lower_bound(l); if (*it != l){
        ptr -= 1; it = lin.lower_bound(l); ptr += 1;
    }
    if (it != lin.begin() && next(it) != lin.end()){
        if (intersect(*prev(it), *next(it)) == 1){ return
        {prev(it)->sc, next(it)->sc}; }
    } lin.erase(it);
} return {-1, -1};
}

pl4 arr[100020]; // (2y, 2x). // (y, x) if fully
nonintersected.
for (int t = 1; t <= 3; t++){
    for (int i = 1; i <= n; i++){
        pl2 p1 = arr[i].fr, p2 = arr[i].sc;
        if (t == 2){ swap(p1.fr, p1.sc); swap(p2.fr, p2.sc); }
        if (t == 3){ p1 = pl2(p1.fr+p1.sc, p1.fr-p1.sc); p2 =
        pl2(p2.fr+p2.sc, p2.fr-p2.sc); }
        if (pl2(p1.sc, p1.fr) > pl2(p2.sc, p2.fr)){ swap(p1, p2);
        } prr[i] = {p1, p2};
        pos[2*i-1] = { {p1.sc, 0}, {p1.fr, i} };
        pos[2*i] = { {p2.sc, -1}, {p2.fr, i} }; // +1 if fully
        nonintersected.
    } pi2 res = solve(n); if (res != pi2(-1, -1)){ cout <<
    "NO"; goto done; }
}

```

```
} cout << "YES";
```

### 7.13 • Faces of a Planar Graph

As always, the polygon are in counterclockwise, Except the outer face which is inner. If you want to include the inner face, remove `val > 0`.

```
p12 arr[100020]; vector<int> adj[100020];
vector<int> chk[100020];
```

```
vector< vector<int> > res;
```

```
vector<int> vtx;
void dfs(int u, int ui){
    chk[u][ui] = 1; if (vtx.size() >= 2 &&
    *prev(prev(vtx.end())) == u){
        vtx.pop_back();
    } else{ vtx.push_back(u); }
    int v = adj[u][ui]; int vi = lower_bound(adj[v].begin(),
    adj[v].end(), u, [&v](int w1, int w2){
        if ((arr[v] < arr[w1]) != (arr[v] < arr[w2])){ return
        (arr[v] < arr[w1]) > (arr[v] < arr[w2]); }
        int res = ccw(arr[v], arr[w1], arr[w2]); return res > 0;
    }) - adj[v].begin(); int vl = adj[v].size();
    vi = (vi == 0 ? vl-1 : vi-1); int w = adj[v][vi];
    int p = (vtx.size() >= 2 ? *prev(prev(vtx.end())) : -1);
    if (!chk[v][vi]){ dfs(v, vi); }
}

for (int v = 1; v <= n; v++){
    chk[v].resize(adj[v].size()); for (int& x : chk[v]){ x = 0;
    }
    sort(adj[v].begin(), adj[v].end(), [&v](int w1, int w2){
        if ((arr[v] < arr[w1]) != (arr[v] < arr[w2])){ return
        (arr[v] < arr[w1]) > (arr[v] < arr[w2]); }
        int res = ccw(arr[v], arr[w1], arr[w2]); return res > 0;
    });
}

for (int v = 1; v <= n; v++){
    int vl = adj[v].size(); for (int vi = 0; vi < vl; vi++){
        if (chk[v][vi]){ continue; }
        vtx.clear(); dfs(v, vi); if (!vtx.empty()){
            int vl = vtx.size(); ll val = 0; for (int i = 0; i <
            vl; i++){
                int j = (i+1 == vl ? 0 : i+1);
                int p1 = vtx[0], p2 = vtx[i], p3 = vtx[j];
                val += area2(arr[p1], arr[p2], arr[p3]);
            } if (val > 0){ res.push_back(vtx); }
        }
    }
}
```

### 7.14 • Polygon Raycast

```
inline ll dot(p12 p1, p12 p2){ return p1.sc*p2.sc +
p1.fr*p2.fr; }
inline ll cross(p12 p1, p12 p2){ return p1.sc*p2.fr -
p1.fr*p2.sc; }
```

```
inline p12 norm(p12 p){
    if (p == p12(0, 0)){ return p; }
    ll g = gcd(p.fr, p.sc); p.fr /= g; p.sc /= g;
    if (p.sc < 0){ p = {-p.fr, -p.sc}; } return p;
} inline p12 norm(ll a, ll b){ return norm({a, b}); }

using pl3 = pair<p12, int>;
vector<pl3> res; // out->line / in->line / line->out /
line->in
// convex(out->out) / concave(in->in) /
out->in / in->out
p12 ray(const vector<p12>& arr, p12 p, p12 d){
    ll g = gcd(d.fr, d.sc); d.fr /= g; d.sc /= g;
    int n = arr.size();
    vector<pl3> tmp; for (int i = 0; i < n; i++){
        int j = (i+1 == n ? 0 : i+1); p12 p1 = arr[i], p2 =
        arr[j];
        p12 v = {p2.fr-p1.fr, p2.sc-p1.sc};
        p12 d1 = {p1.fr-p.fr, p1.sc-p.sc}, d2 = {p2.fr-p.fr,
        p2.sc-p.sc};
        int s1 = sgn(cross(d, d1)), s2 = sgn(cross(d, d2));
        if (s1 == 0){ tmp.push_back({norm(dot(d, d1), dot(d, d)),
        s2}); }
        if (s2 == 0){ tmp.push_back({norm(dot(d, d2), dot(d, d)),
        s1}); }
        if (s1*s2 == -1){ tmp.push_back({norm(cross({-d1.fr,
        -d1.sc}, v), cross(v, d)), 6}); }
    } sort(tmp.begin(), tmp.end(), [](const pl3& p1, const pl3&
    p2){
        p12 f1 = p1.fr, f2 = p2.fr; return (i128)f1.fr*f2.sc <
        (i128)f2.fr*f1.sc;
    }); res.clear();
    int l = tmp.size(); for (int i = 0; i < l; i++){
        if (i+1 < l && tmp[i].fr == tmp[i+1].fr && tmp[i].sc !=
        6){
            int s1 = tmp[i].sc, s2 = tmp[i+1].sc;
            res.push_back({tmp[i].fr, s1*s2==0 ? (1-s1-s2)/2 :
            s1*s2 > 0 ? 4 : 6}); i += 1;
        } else{ res.push_back(tmp[i]); }
    }
    int pos = 0; // out / in / line ccw / line cw
    for (pl3& p : res){
        if (p.sc == 4){ p.sc ^= pos; }
        else if (p.sc == 6){ p.sc ^= pos; pos ^= 1; }
        else if (p.sc == 0){ p.sc = pos; pos ^= 2; }
        else if (p.sc == 1){ p.sc = pos^(pos>>1); pos ^= 3; }
    } return {g, 1};
}
```

```
bool fcmp(p12 p1, p12 p2){ return (i128)p1.fr*p2.sc <
(i128)p2.fr*p1.sc; }
// Assuming that p1 and p2 are not inside; tangent point/line
permitted
bool visible(const vector<p12>& arr, p12 p1, p12 p2){
    if (p1 == p2){ return 1; }
```

```
p12 st = {0, 1}, ed = ray(arr, p1, {p2.fr-p1.fr,
p2.sc-p1.sc});
vector<p12> pos; for (pl3 e : res){
    int ei = e.sc; //cout << "event " << e.fr.fr << '/' <<
    e.fr.sc << ' ' << e.sc << endl << flush;
    if (ei == 1 || ei == 5 || ei == 7){ pos.push_back(e.fr);
    }
    if (ei == 3 || ei == 5 || ei == 6){ pos.push_back(e.fr);
    }
} int el = pos.size(); //cout << el << endl << flush;
for (int i = 0; i < el; i+=2){
    p12 f1 = pos[i], f2 = pos[i+1];
    if (fcmp(st, f2) && fcmp(f1, ed)){ return 0; }
} return 1;
}
```

## 8 Graph Theory - Distance & Connectivity

### 8.1 • Eulerian Path & Cycle

Directed version. For undirected use  $e_i = (v_i, w_i)$  and save  $i$  like BCC.

```
vector<int> adj[200020]; int ptr[200020];
int pth[200020]; int len;
void dfs(int now){
    pth[++len] = now;
    for (int& p : ptr[now]; p < adj[now].size();){
        int nxt = adj[now][p++];
        dfs(nxt);
    }
}
```

### 8.2 • Strongly Connected Component

- Time Complexity:  $\mathcal{O}(V + E)$ ..

```
stack<int> stk; int scc[N+20];
int ord[N+20], ont;
int dfs(int now){
    stk.push(now); int res = ord[now] = ++ont;
    for (int nxt : adj[now]){
        if (ord[nxt] == 0){ res = min(res, dfs(nxt)); }
        else if (scc[nxt] == 0){ res = min(res, ord[nxt]); }
    }
    if (res == ord[now]){
        while (!stk.empty()){
            int vtx = stk.top(); stk.pop();
            scc[vtx] = now; if (vtx == now){ break; }
        }
    } return res;
}
```

### 8.3 • Biconnected Component

- Time Complexity:  $\mathcal{O}(V + E)$ ..

```
pi3 edg[100020]; // {v, w, color}
vector<int> adj[100020]; // Edge Index
stack<int> stk;
```

```
int ord[100020], ont = 0;
int dfs(int now, int pre){
    int res = ord[now] = ++ont;
    for (int ei : adj[now]){
        int nxt = edg[ei].fr.fr ^ edg[ei].fr.sc ^ now;
        if (nxt == pre){ continue; }
        if (ord[now] > ord[nxt]){ stk.push(ei); }
        if (ord[nxt] > 0){ res = min(res, ord[nxt]); }
        else{
            int val = dfs(nxt, now); res = min(res, val);
            if (val >= ord[now]){
                vector<int> v; while (!stk.empty()){
                    int e = stk.top(); stk.pop();
                    edg[e].sc = ei; if (e == ei){ break; }
                }
            }
        }
    }
    return res;
}
```

### Articulation Point & Edge

- Articulation Point: Vertex with 2 or more colors.
- Articulation Edge: Edge that has unique color.

## 8.4 • Closure Problem

Given a directed acyclic graph  $G = (V, E)$ , with each vertex  $v$  having weight  $w_v$ , maximize the sum of the weight of the selected vertices such that if vertex  $v$  were chosen, then you also have to choose the vertex  $w$  if  $v \rightarrow w$ .

This can be modeled as following. Create a source  $s$ , sink  $t$ .

- If  $w_v \geq 0$ , then  $s \rightarrow v$  with weight  $w_v$ .
- If  $w_v \leq 0$ , then  $v \rightarrow t$  with weight  $-w_v$ .
- If  $v \rightarrow w$ , then  $v \rightarrow w$  with weight  $\infty$ .

Find the minimum cut, then select only the vertices that are in  $s$ -side. The sum itself is the  $\sum_{w_v \geq 0} w_v - \text{MaxFlow}$ .

## 9 Graph Theory - Network Flow & Matching

### 9.1 • Relations between Flow and Matching

On a Bipartite Graph, Size of Minimum Vertex Cover is equal to the size of Maximum Matching.

Size of Maximum Independent Set is  $V$  - size of Minimum Vertex Cover.

### 9.2 • Diniz

```
int n, m; int src, snk, nc; ll adj[220][220];
int dis[220];
bool bfs(int st, int ed){
    memset(dis, -1, sizeof(dis)); dis[st] = 0;
    queue<int> q; q.push(st); while (!q.empty()){
        int now = q.front(); q.pop();
        for (int nxt = 0; nxt < nc; nxt++){
            if (adj[now][nxt] == 0){ continue; }
            if (dis[nxt] != -1){ continue; }
        }
    }
}
```

```
dis[nxt] = dis[now]+1; q.push(nxt);
}
} return dis[ed] != -1;
}
bool chk[220]; int ptr[220];
int pth[220], pc;
bool dfs(int now, int idx){
    pth[idx] = now; chk[now] = 1; if (now == snk){ pc = idx;
    return 1; }
    for (int& nxt = ptr[now]; nxt < nc; nxt++){
        if (adj[now][nxt] == 0){ continue; }
        if (dis[now]+1 != dis[nxt]){ continue; }
        if (chk[nxt]){ continue; }
        if (dfs(nxt, idx+1)){ return 1; }
    }
    return 0;
}

memset(ptr, 0, sizeof(ptr)); memset(chk, 0, sizeof(chk));
ll ans = 0; while (bfs(src, snk)){
    memset(ptr, 0, sizeof(ptr)); memset(chk, 0, sizeof(chk));
    while (dfs(src, 0)){
        memset(chk, 0, sizeof(chk));
        ll res = INF; for (int i = 1; i <= pc; i++){
            int v = pth[i-1], w = pth[i];
            res = min(res, adj[v][w]);
        }
        ans += res;
        for (int i = 1; i <= pc; i++){
            int v = pth[i-1], w = pth[i];
            adj[v][w] -= res; adj[w][v] += res;
        }
    }
}
```

### 9.3 • Hungarian Algorithm

```
// Max version. for Min, adj[v][w] *= -1, then print -ans.
ll adj[520][520];
ll l1[520], l2[520]; ll d[520]; int di[520];
bool c1[520], c2[520]; int m1[520], m2[520];
int par[520];
inline ll cost(int v, int w){ return l1[v] + l2[w] -
adj[v][w]; }
void psh(int n, int v, int p){ // p -> m1[v] -> v
    c1[v] = 1; par[v] = p;
    for (int j = 1; j <= n; j++){
        if (cost(v, j) < d[j]){ d[j] = cost(v, j); di[j] = v; }
    }
}

for (int i = 1; i <= n; i++){ l1[i] = *max_element(adj[i]+1,
adj[i]+n+1); }
for (int j = 1; j <= n; j++){ l2[j] = 0; }
for (int cnt = 0; cnt < n; cnt++){
    memset(c1, 0, sizeof(c1)); memset(c2, 0, sizeof(c2));
    memset(par, 0, sizeof(par));
    queue<int> q; int st = 0;
    for (int i = 1; i <= n; i++){
        if (m1[i] == 0){
```

```
st = i; q.push(i);
        par[i] = -1; c1[i] = 1; break;
    }
}
for (int j = 1; j <= n; j++){ d[j] = cost(st, j); di[j] =
st; }
int v = 0, w = 0; while (1){
    while (!q.empty()){
        v = q.front(); q.pop();
        for (int j = 1; j <= n; j++){
            if (cost(v, j) == 0 && !c2[j]){
                if (m2[j] == 0){ w = j; goto augment; }
                c2[j] = 1; q.push(m2[j]); psh(n, m2[j], v);
            }
        }
    }
    ll val = INF; for (int j = 1; j <= n; j++){
        if (!c2[j]){ val = min(val, d[j]); }
    }
    for (int i = 1; i <= n; i++){ if (c1[i]){ l1[i] -= val; }
    }
    for (int j = 1; j <= n; j++){
        if (c2[j]){ l2[j] += val; } else{ d[j] -= val; }
    }
    while (!q.empty()){ q.pop(); }
    for (int j = 1; j <= n; j++){
        if (!c2[j] && d[j] == 0){
            if (m2[j] == 0){ v = di[j]; w = j; goto augment; }
            else{
                c2[j] = 1; if (!c1[m2[j]]){
                    q.push(m2[j]); psh(n, m2[j], di[j]);
                }
            }
        }
    }
}
} augment:
while (v != -1){
    int pv = m2[w], pw = m1[v];
    m1[v] = w; m2[w] = v; v = par[v]; w = pw;
}
}
ll ans = 0; for (int i = 1; i <= n; i++){ ans +=
adj[i][m1[i]]; }
```

### 9.4 • Minimum Cost Maximum Flow

```
int src, snk, nc; pi2 adj[220][220]; // (flow, cost)
int dis[220]; bool chk[220]; int pre[220];
int pth[220]; int pc;
bool spfa(int st, int ed){
    memset(dis, 0x3f, sizeof(dis)); memset(chk, 0,
sizeof(chk));
    queue<int> q; dis[st] = 0; chk[st] = 1; q.push(st);
    while (!q.empty()){
        int now = q.front(); q.pop(); chk[now] = 0;
        for (int nxt = 1; nxt <= nc; nxt++){
```

```

    if (adj[now][nxt].fr == 0){ continue; }
    if (dis[nxt] <= dis[now] + adj[now][nxt].sc){ continue; }
    }
    dis[nxt] = dis[now] + adj[now][nxt].sc; pre[nxt] = now;
    if (!chk[nxt]){ chk[nxt] = 1; q.push(nxt); }
}
}
if (dis[ed] == INF){ return 0; }
int ptr = ed; pc = 0;
while (ptr != st){ pth[pc++] = ptr; ptr = pre[ptr]; }
pth[pc] = ptr;
reverse(pth, pth+pc+1); return 1;
}

int ans = 0, res = 0; while (spfa(src, snk)){
    int cnt = INF; for (int i = 1; i <= pc; i++){
        int v = pth[i-1], w = pth[i];
        cnt = min(cnt, adj[v][w].fr);
    } res += cnt;
    for (int i = 1; i <= pc; i++){
        int v = pth[i-1], w = pth[i];
        adj[v][w].fr -= cnt; adj[w][v].fr += cnt;
        ans += cnt*adj[v][w].sc;
    }
}

```

## 9.5 • Minimum Path Cover

Given a DAG, Minimum Vertex Path Cover can be computed as follow. Divide the vertex  $v$  onto  $v_l$  and  $v_r$ . For each edge  $v \rightarrow w$ , connect  $v_l$  and  $w_r$ . Number of path you need to use is the  $V$  - Maximum Matching.

Minimum Edge Path Cover is just Eulerian Path in disguise.

## 9.6 • Hall's Marriage Theorem

A bipartite graph  $G = (X + Y, E)$  has a perfect  $X$ -matching iff for all subset  $V \subseteq X$ ,  $|V| \leq |N_G(V)|$ .

A bipartite graph  $G = (X + Y, E)$  have a maximum  $X$ -matching of size  $|X| - \max_{V \subseteq X} |V| - |N_G(V)|$ .

## 9.7 • Stable Marriage Problem

For  $i$  in 1 to  $N$ , try to match the best possible matching  $j$ , knocking other matchings in process.

This matching is best for proposer  $i$  while worst for proposee  $j$ .

```

int adj[1020][1020]; int rnk[1020][1020]; // adj[i][r] -> j
// rnk[j][i] -> r
int ptr[1020]; int res[1020]; // ptr[i] = r // res[j] = i
bool f(int i, int j){
    if (res[j] == 0){ res[j] = i; return 1; }
    int p = res[j]; if (rnk[j][p] > rnk[j][i]){
        res[j] = i; ptr[p] += 1; for (int& r = ptr[p]; ; r++){
            if (f(p, adj[p][r])){ break; }
        } return 1;
    } return 0;
}

for (int i = 1; i <= n; i++){ ptr[i] = 1; }

```

```

for (int i = 1; i <= n; i++){
    for (int& r = ptr[i]; ; r++){
        if (f(i, adj[i][r])){ break; }
    }
}

```

## 9.8 • Flow with Demands

- $req_i$  is required flow  $i$ th vertex need to send
  - $req_i < 0$ : Need to get it from source
  - $req_i > 0$ : Need to send it to sink
- $v \rightarrow w$  with lower/upper bound of  $l$  and  $u$ .
  - $req_v = l$ ;  $req_w = -l$
  - Capacity of  $v \rightarrow w = u - l$
- Create new source and sink, and connect the  $v$  with source or sink with capacity  $|req_v|$ .
- From the old sink to old source, connect an edge with the capacity  $\infty$ . If there were none, ignore this.
- MaxFlow should be equal to  $\sum_{req_v > 0} req_v$  in order to satisfy the original graph.

## 9.9 • General Matching

```

int n, m; vector<int> adj[520]; // Input: Graph
int res[520]; // Matched Vertex

int par[520]; // BFS path, before v.
int chk[520]; // Visited? (0: no, 1: yes+odd, 2: yes+even) \
               odd/even comes from bipartite... kinda. \
               all v in q satisfies chk[v] = 1.
void pro(int r, int v){ // Re-Match. a.k.a. Augmenting
    int p = v; do{
        p = par[v]; int q = res[p];
        res[v] = p; res[p] = v; v = q;
    } while (r != p);
}

int bls[520]; // Position of a Blossom
bool vst[520]; // Used for lca.
int lca(int v, int w){ // LCA on a BFS tree.
    memset(vst, 0, sizeof(vst));
    while (1){
        if (v != 0){
            if (vst[v]){ return v; }
            vst[v] = 1; v = bls[par[res[v]]];
        } swap(v, w);
    }
}

void cyc(int v, int w, queue<int>& q){ // Cycle (fancy term = Blossom) Merging.
    int l = lca(bls[v], bls[w]);
    while (bls[v] != 1){
        par[v] = w; w = res[v];
        if (chk[w] == 2){ q.push(w); chk[w] = 1; }
        bls[v] = bls[w] = 1; v = par[w];
    }
}

```

```

}
}
bool bfs(int r){ // main BFS.
    memset(chk, 0, sizeof(chk)); memset(par, 0, sizeof(par));
    for (int v = 1; v <= n; v++){ bls[v] = v; }
    queue<int> q; q.push(r); chk[r] = 1; while (!q.empty()){
        int v = q.front(); q.pop();
        for (int w : adj[v]){
            if (chk[w] == 0){
                par[w] = v; chk[w] = 2;
                if (res[w] == 0){ pro(r, w); return 1; }
                q.push(res[w]); chk[res[w]] = 1;
            }
            else if (chk[w] == 1 && bls[v] != bls[w]){
                cyc(w, v, q); cyc(v, w, q);
            }
        }
    } return 0;
}

void Main(){
    cin >> n >> m; while (m--){
        int v, w; cin >> v >> w;
        adj[v].push_back(w); adj[w].push_back(v);
    }
    int ans = 0; for (int v = 1; v <= n; v++){
        if (res[v] == 0 && bfs(v)){ ans += 1; }
    } cout << ans;
}

```

## 10 Graph Theory - Miscellaneous

### 10.1 • Tree Isomorphism

```

int dep[2][100020]; vector<int> vtx[2][100020]; int
par[2][100020];
int dfs(int t, int v, int p){
    dep[t][v] = dep[t][p]+1; vtx[t][dep[t][v]].push_back(v);
    par[t][v] = p; erase(adj[t][v], p);
    int res = dep[t][v]; for (int w : adj[t][v]){
        if (w == p){ continue; }
        res = max(res, dfs(t, w, v));
    } return res;
}

int val[2][100020]; int ans[100020];
using pvi = pair<vector<int>, int>;

centroid(0, 1); pi2 c0 = cent; centroid(1, 1); pi2 c1 = cent;
if ((c0.sc == -1) != (c1.sc == -1)){ cout << "EI"; return; }
for (int t = 0; t < 2; t++){
    pi2& c = (t==0 ? c0 : c1); if (c.sc == -1){ continue; }
    int v = c.fr, w = c.sc; erase(adj[t][v], w);
    erase(adj[t][w], v);
    adj[t][v].push_back(n+1); adj[t][w].push_back(n+1);
    adj[t][n+1].push_back(v); adj[t][n+1].push_back(w);
    c = {n+1, 0};
}

```



```

}
int d0 = dfs(0, c0.fr, c0.fr); int d1 = dfs(1, c1.fr, c1.fr);
if (d0 != d1){ cout << "EI"; return; }
for (int d = d0; d > 1; d--){
    vector<pvi> res[2];
    for (int t = 0; t < 2; t++){
        for (int v : vtx[t][d-1]){
            res[t].emplace_back(); res[t].back().sc = v;
            for (int w : adj[t][v]){
                res[t].back().fr.push_back(val[t][w]); }
        } for (pvi& p : res[t]){ sort(p.fr.begin(), p.fr.end()); }
        sort(res[t].begin(), res[t].end());
    }
}
int l0 = res[0].size(), l1 = res[1].size(); if (l0 != l1){
    cout << "EI"; return; }
int num = 0; for (int i = 0; i < l0; i++){
    if (res[0][i].fr != res[1][i].fr){ cout << "EI"; return; }
    if (i > 0 && res[0][i-1].fr != res[0][i].fr){ num += 1; }
    int v0 = res[0][i].sc, v1 = res[1][i].sc;
    val[0][v0] = val[1][v1] = num;
}
}
queue<pi2> q; q.push({c0.fr, c1.fr}); while (!q.empty()){
    int r0 = q.front().fr, r1 = q.front().sc; q.pop(); ans[r0]
    = r1;
    sort(adj[0][r0].begin(), adj[0][r0].end(), [](int v1, int
    v2){ return val[0][v1] < val[0][v2]; });
    sort(adj[1][r1].begin(), adj[1][r1].end(), [](int v1, int
    v2){ return val[1][v1] < val[1][v2]; });
    int l = adj[0][r0].size(); for (int i = 0; i < l; i++){
        q.push({adj[0][r0][i], adj[1][r1][i]}); }
}
cout << "JAH" << endl; for (int i = 1; i <= n; i++){ cout <<
ans[i]-1 << endl; }

```

## 10.2 • Graph Realization Problem

Simple Undirected Graph with Degree Sequence  $d_1 \geq d_2 \geq \dots \geq d_n$  exists iff  $\sum_i d_i$  is even and  $\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$  holds for every  $1 \leq k \leq n$ .

You can construct the graph by picking  $d_1$  and connecting it to  $2, 3, \dots, d_{d_1+1}$  and pass everything else as induction. Note that  $d_2, d_3, \dots, d_{d_1+1}$  also decreases by 1 by doing this, and thus might need reordering. You can make the Connected Graph by picking  $d_n$  and connecting it to  $1, 2, \dots, d_n$ .

Simple Directed Graph with In/Outdegree Sequence  $(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n)$  with  $a$  being nonincreasing exists iff  $\sum_i a_i = \sum_i b_i$  and  $\sum_{i=1}^k a_i \leq \sum_{i=1}^k \min(b_i, k-1) + \sum_{i=k+1}^n \min(b_i, k)$ .

You can construct the graph by picking max  $b_i$  and connecting it to vertices with max  $a_j$ s. You can make the Weakly Connected Graph by picking  $\min(a_i + b_i)$ , and connecting  $a_i$  to max  $b_j$ , ties broken by  $a_j$ , then connecting  $b_i$  to max  $a_j$ , ties broken by  $b_j$ . Note that the degree sequence have to be sorted before connecting  $b_i$  as well for the tiebreaks.

Simple Bipartite Graph with Degree sequence  $a_1 \geq a_2 \geq \dots \geq a_n$

and  $b_1 \geq b_2 \geq \dots \geq b_m$  exists iff  $\sum_i a_i = \sum_j b_j$  and  $\sum_{i=1}^k a_i \leq \sum_{i=1}^m \min(b_i, k)$  for all  $1 \leq k \leq n$ .

You can construct the graph by greedily matching  $a_i$  with max  $b_j$ s, for  $i$  in 1 to  $n$ .

## 11 Dynamic Programming

### 11.1 • Monge Array

Monge (intersection is better than containment): CHT, DnC opt, WQS, MonoQ, Knuth.

Let  $C_{i,j} = A_i + B_j$ . If  $A$  or  $B$  is convex (i.e. Slope is Increasing), then  $C$  is monge.

### 11.2 • Rerooting

Tree DP, with changing root. For all direction: Edge  $v \rightarrow w$ , set root as  $v$  and record subtree  $w$ .

- Time Complexity:  $\mathcal{O}(N)$ ..

```

void reroot(int now, int pre){
    ans[now] = dp[now]; for (int nxt : adj[now]){
        if (nxt == pre){ continue; }
        int dpv = dp[now], dpw = dp[nxt];
        dp[now] -= (dp[nxt]?); dp[nxt] += (dp[now]?);
        reroot(nxt, now);
        dp[now] = dpv; dp[nxt] = dpw;
    }
} // dpf(root, root); reroot(root, root)

```

### 11.3 • Sum over Subset

$$D_S = \sum_{T \subseteq S} A_T.$$

- Time Complexity:  $\mathcal{O}(N2^N)$ ..

```

for (int i = 0; i < X; i++){ dp[i] = arr[i]; }
for (int b = 0; b < B; b++){
    for (int bit = 0; bit < X; bit++){
        if (bit >> b & 1){ dp[bit] += dp[bit ^ 1 << b]; }
    }
}

```

### 11.4 • Convex Hull Trick

$$D_i = \max_{j < i} a_j x_i + b_j.$$

- Time Complexity:  $\mathcal{O}(N)$  for  $N$  insertions,  $\mathcal{O}(\log N)$  for query..

```

inline ld crs(p12 l1, p12 l2){ return
(ld)(l2.sc-l1.sc)/(l1.fr-l2.fr); }
pair<p12, ld> stk[1000020]; int sp = 0;
inline void psh(p12 p){
    stk[sp].fr = p; stk[sp].sc = (sp==0 ? 0 : crs(stk[sp-1].fr,
    stk[sp].fr));
    sp += 1;
}
inline void pop(){ sp -= 1; }

```

```

// query: x
int st = 0, ed = sp; while (st+1 <= ed-1){
    int mid = st+ed >> 1;

```

```

    if (x < stk[mid].sc){ ed = mid; } else{ st = mid; }
} p12 p = stk[st].fr; // A(j)*x + B(j)
dp[i] = p.fr*x + p.sc + C(i);
// update: p = (A(j), B(j))
while (sp > 0){
    if (crs(stk[sp-1].fr, p) <= stk[sp-1].sc){ pop(); } else{
        break; }
} psh(p);

```

### 11.5 • Divide and Conquer Optimization

$$D_{k,i} = \max_j < i D_{k-1,j} + A_{j,i}. \text{ opt}_i < \text{opt}_{i+1}.$$

- Time Complexity:  $\mathcal{O}(KN \log N)$ ..

```

ll dp[2][N+20]; // Sliding Window on first index
void dnc_opt(ll* dp1, ll* dp0, int is, int ie, int os, int
oe){
    if (is > ie){ return; } int im = is+ie >> 1;
    int o = os; for (int p = os; p <= min(oe, im-1); p++){
        if (f(dp0, p, im) < f(dp0, o, im)){ o = p; }
    } dp1[im] = f(dp0, o, im);
    dnc_opt(dp1, dp0, is, im-1, os, o); dnc_opt(dp1, dp0, im+1,
    ie, o, oe);
}

```

### 11.6 • WQS Binary Search

$$D_{k,i} = \min_{j < i} D_{k-1,j} + A_{j,i}. D_{k,N} \text{ must be convex.}$$

- Time Complexity:  $\mathcal{O}(T(N) \log X)$ ..

```

// solve(n, mid) returns the number of used elements
// Binary Search over Lambda
ll st = 0, ed = 1e18; ll ans = 1e18;
while (st <= ed){
    ll mid = st+ed >> 1;
    int cnt = solve(n, mid); ans = min<i128>(ans, dp[n] +
    (i128)mid*k);
    if (cnt <= k){ ed = mid-1; } else{ st = mid+1; }
} cout << ans;

```

### 11.7 • Monotone Queue Optimization

$D_i = \min_{j < i} D_j + A_{j,i}$ . For all pair  $i < j$ , there exists a unique point  $p$  that  $D_i + C_{i,p}$  and  $D_j + C_{j,p}$  changes. So  $k < p$  means  $<$ , and  $\geq$  otherwise.

- Time Complexity:  $\mathcal{O}(N \log N)$ .

Code from BOJ 17439 - Flower Shop.

```
ll arr[50020]; ll prf[50020];
```

```
deque<int> dq;
pl2 dp[50020];
inline ll f(int j, int i){ return (prf[i]-prf[j])*(i-j); }
inline ll dpf(int j, int i){ return dp[j].fr + f(j, i); }
inline int crx(int a, int b, int n){ // Assume a < b.
    int st = b+1, ed = n; int i = n+1; while (st <= ed){
        int mid = st+ed >> 1;
        if (dpf(a, mid) >= dpf(b, mid)){ i = mid; ed = mid-1; }
        else{ st = mid+1; }
    } return i;
}
int solve(int n, ll x){
    dp[0] = {0, 0}; for (int i = 1; i <= n; i++){
        while (dq.size() >= 2){
            int dql = dq.size();
            int a = dq[dql-2], b = dq[dql-1];
            if (crx(a, b, n) >= crx(b, i-1, n)){ dq.pop_back(); }
            else{ break; }
        } dq.push_back(i-1);
        while (dq.size() >= 2){
            int a = dq[0], b = dq[1];
            if (crx(a, b, n) <= i){ dq.pop_front(); }
            else{ break; }
        } int j = dq.front();
        dp[i] = {dpf(j, i), dp[j].sc+1};
        //cout << dp[i].fr << " \n"[i==n];
    } return dp[n].sc;
}
```

## 11.8 Slope Trick

Let  $f$  be a convex piecewise linear function with integer coefficient. We can save this function by the points where the line changes, with one linear function.

We can add two functions by merging the points without removing duplicants, and adding the linear function. Use `priority_queue` if needed.

## 11.9 Knuth Optimization

$$D_{i,j} = D_{i,k} + D_{k+1,j} + A_{i,j} \text{ where } \text{opt}_{i,j-1} \leq \text{opt}_{i,j} \leq \text{opt}_{i+1,j}.$$

- Time Complexity:  $\mathcal{O}(N^2)$ .

```
for (int d = 0; d < n; d++){
    for (int s=1, e=1+d; e <= n; s++, e++){
        if (d == 0){ dp[s][e] = 0; opt[s][e] = s; }
        else{
            opt[s][e] = s; dp[s][e] = INF;
            for (int p = opt[s][e-1]; p <= opt[s+1][e]; p++){
                ll res = dp[s][p] + dp[p+1][e] + prf[e]-prf[s-1];
                if (dp[s][e] > res){ dp[s][e] = res; opt[s][e] = p; }
            }
        }
    }
}
cout << dp[1][n] << endl;
```

## 11.10 Kitamasu

$A_n = \prod_{i=1}^k A_{n-d} C_d$ . The code below uses `arr` as initial terms with index 0 to  $n-1$ . And `f` as coefficient of  $x^n - \sum_{d=1}^k C_d x^{n-d}$ .

```
const ll mod = 104857601; const ll r = 3;

ll fpow(ll mul, ll bit){
    ll res = 1; while (bit){
        if (bit&1){ res = res*mul % mod; }
        mul = mul*mul % mod; bit >>= 1;
    } return res;
} inline ll finv(ll x){ return fpow(x, mod-2); }

inline void normp(vector<ll>& arr){
    int l = arr.size(); while (l > 0){
        if (arr[l-1] == 0){ l -= 1; } else{ break; }
    } arr.resize(l);
}

inline vector<ll> trim(vector<ll> arr, int mx){
    int l = min<int>(arr.size(), mx); return
    vector<ll>(arr.begin(), arr.begin()+l);
}

void dft(vector<ll>& arr, bool inv = false){
    int n = arr.size();
    for (int j=0, i=1; i < n; i++){
        int bit = n>>1;
        while (j&bit){ j ^= bit; bit >>= 1; } j ^= bit;
        if (i < j){ swap(arr[i], arr[j]); }
    }
    for (int l = 1; l < n; l*=2){
        ll w = fpow(r, (mod-1)/(2*l)); if (inv){ w = finv(w); }
        for (int i = 0; i < n; i += l*2){
            ll wp = 1; for (int j = 0; j < l; j++){
                ll a = arr[i+j], b = arr[i+j+l]*wp % mod;
                arr[i+j] = (a+b) % mod; arr[i+j+l] = (a-b+mod)%mod;
                wp = wp*w % mod;
            }
        }
    }
    if (inv){
        ll invn = finv(n);
        for (int i = 0; i < n; i++){ arr[i] = arr[i]*invn % mod; }
    }
}

vector<ll> mulp(vector<ll> arr, vector<ll> brr){ // A \times B
    normp(arr); normp(brr);
    int al = arr.size(), bl = brr.size();
    int n = max(al, bl); n = 1 << (bits(n) + 1 - ((n&n)==n));
    arr.resize(n); brr.resize(n);
    dft(arr); dft(brr);
    for (int i = 0; i < n; i++){ arr[i] = arr[i]*brr[i] % mod; }
    dft(arr, -1); arr = trim(arr, al+bl-1); return arr;
}
```

```
vector<ll> invp(vector<ll> arr, int k){ // A^{-1} \pmod{x^k}.
    Assume A[0] \neq 0.
    arr = trim(arr, k);
    vector<ll> res = vector<ll>{finv(arr[0])}; int m = 1;
    while (m < k){
        vector<ll> val = trim(mulp(trim(arr, m*2), res), m*2);
        int vl = val.size();
        for (int i = 0; i < vl; i++){
            val[i] = ((i==0)*2 + mod-val[i]) % mod;
        } res = trim(mulp(res, val), m*2); m *= 2;
    } res = trim(res, k); return res;
}

vector<ll> divp(vector<ll> arr, vector<ll> brr){ // D, where
    A = BQ + R with \deg R < \deg B.
    normp(arr); normp(brr);
    int al = arr.size(), bl = brr.size(); if (al < bl){ return
    vector<ll>{}; }
    reverse(arr.begin(), arr.end()); reverse(brr.begin(),
    brr.end());
    arr = trim(arr, al-bl+1); brr = trim(brr, al-bl+1);
    vector<ll> qrr = mulp(arr, invp(brr, al-bl+1));
    qrr.resize(al-bl+1);
    reverse(qrr.begin(), qrr.end()); normp(qrr); return qrr;
}

vector<ll> modp(vector<ll> arr, vector<ll> brr){ // R, where
    A = BQ + R with \deg R < \deg B.
    normp(arr); normp(brr);
    int al = arr.size(), bl = brr.size(); if (al < bl){ return
    arr; }
    vector<ll> qrr = divp(arr, brr); vector<ll> drr = mulp(brr,
    qrr);
    for (int i = 0; i < al; i++){ arr[i] = (arr[i]-drr[i]+mod)
    % mod; }
    normp(arr); return arr;
}

ll arr[30020], crr[30020];

ll kth(ll k, const vector<ll>& arr, const vector<ll>& f){ //
    x^k \pmod f
    int n = arr.size(); if (k < n){ return arr[k]; }
    vector<ll> res{1}, mul{0, 1}; while (k){
        if (k&1){ res = modp(mulp(res, mul), f); }
        mul = modp(mulp(mul, mul), f); k >>= 1;
    }
    ll ans = 0; for (int i = 0; i < n; i++){ ans +=
    res[i]*arr[i] % mod; }
    return ans%mod;
}
```

## 11.11 Hirschberg

Unlike other string algorithms, this assumes that `s1` and `s2` is 1-indexed.

```

string s1, s2;

int dp[3][7010];
void solve(int i1, int j1, int i2, int j2, bool top = true){
    if (i1 > j1 || i2 > j2){ return; }
    if (i1 == j1){
        for (int p2 = i2; p2 <= j2; p2++){
            if (s1[i1] == s2[p2]){
                if (top){ cout << 1 << endl; }
                cout << s1[j1] << endl; return;
            }
        }
        if (top){ cout << 0 << endl; } return;
    }
    int m1 = i1+j1 >> 1;
    for (int p1 = i1-1; p1 <= m1; p1++){
        int b1 = p1&1; int b0 = 1-b1;
        for (int p2 = i2-1; p2 <= j2; p2++){
            if (p1 == i1-1 || p2 == i2-1){ dp[b1][p2] = 0; }
            else{
                dp[b1][p2] = max(dp[b0][p2], dp[b1][p2-1]);
                if (s1[p1] == s2[p2]){ dp[b1][p2] = max(dp[b1][p2],
                    dp[b0][p2-1+1]); }
            }
        }
    }
    if ((m1&1) == 0){
        for (int p2 = i2-1; p2 <= j2; p2++){ dp[1][p2] =
            dp[0][p2]; }
    }
    for (int p1 = j1; p1 > m1; p1--){
        int b1 = p1&1; int b0 = 1-b1; b1 <= 1; b0 <= 1;
        for (int p2 = j2; p2 >= i2-1; p2--){
            if (p1 == j1 || p2 == j2){ dp[b1][p2] = 0; }
            else{
                dp[b1][p2] = max(dp[b0][p2], dp[b1][p2+1]);
                if (s1[p1+1] == s2[p2+1]){ dp[b1][p2] =
                    max(dp[b1][p2], dp[b0][p2+1+1]); }
            }
        }
    }
    if ((m1+1&1) == 0){
        for (int p2 = i2-1; p2 <= j2; p2++){ dp[2][p2] =
            dp[0][p2]; }
    }
    int mx = 0; pi2 mpx = {0, 0}; for (int p2 = i2-1; p2 <= j2;
    p2++){
        int x = dp[1][p2]+dp[2][p2];
        if (x > mx){ mx = x; mpx = {0, p2}; }
    }
    for (int p2 = i2; p2 <= j2; p2++){
        if (s1[m1+1] == s2[p2]){
            int x = dp[1][p2-1]+dp[2][p2]+1;
            if (x > mx){ mx = x; mpx = {1, p2}; }
        }
    }
    if (top){ cout << mx << endl; }
}

```

```

if (mx == 0){ return; }
if (m1 <= 0){
    int p2 = m1.sc;
    solve(i1, m1, i2, p2, 0); solve(m1+1+1, j1, p2+1, j2, 0);
}
else{
    int p2 = m1.sc;
    solve(i1, m1, i2, p2-1, 0); cout << s2[p2]; solve(m1+1+1,
    j1, p2+1, j2, 0);
}
}
}

```

```

void Main(){
    cin >> s1 >> s2; int l1 = s1.size(), l2 = s2.size();
    s1 = " " + s1; s2 = " " + s2;
    solve(1, l1, 1, l2);
}

```

### 11.12 • Connection Profile

Single Connected Region, with DP condition.

- Time Complexity:  $\mathcal{O}(NMB_M)$ ,  $B_9 = 21\,147$ . Actual state is about 2000?

```

int n, m; int arr[10][10];
inline int f(string& bit){
    int cvt[10] = {}; int val = 0;
    for (char& c : bit){
        int x = c-'0'; if (x == 0){ continue; }
        if (cvt[x] == 0){ cvt[x] = ++val; }
        c = cvt[x]+'0';
    } return val;
}
const ll INF = 1e18;
map<string, ll> dp[10][10];
ll dpf(int y, int x, string bit){ int cnt = f(bit);
    if (x == m){ y += 1; x = 0; }
    if (y == n){ return cnt <= 1 ? 0 : INF; }
    if (dp[y][x].count(bit)){ return dp[y][x][bit]; }
    dp[y][x][bit] = INF;
    bool flg = bit.front() == '0';
    for (int b = 1; b < m; b++){ flg |= bit[0] == bit[b]; }
    if (flg){
        string nxt = bit.substr(1, m-1) + '0';
        dp[y][x][bit] = min(dp[y][x][bit], dpf(y, x+1, nxt));
    }
    char u = bit.front(), l = bit.back(); if (x == 0){ l = '0'; }
    if (u == '0' && l == '0'){
        string nxt = bit.substr(1, m-1) + (char)(cnt+1+'0');
        dp[y][x][bit] = min(dp[y][x][bit], dpf(y, x+1, nxt) +
        arr[y][x]);
    }
    else if (u == '0' || l == '0'){
        string nxt = bit.substr(1, m-1) + (u=='0' ? l : u);
        dp[y][x][bit] = min(dp[y][x][bit], dpf(y, x+1, nxt) +
        arr[y][x]);
    }
}

```

```

else{
    string nxt = bit.substr(1, m-1) + u;
    for (char& c : nxt){ if (c == 1){ c = u; } }
    dp[y][x][bit] = min(dp[y][x][bit], dpf(y, x+1, nxt) +
    arr[y][x]);
}
if (cnt <= 1){ dp[y][x][bit] = min<ll>(dp[y][x][bit], 0); }
return dp[y][x][bit];
}
dpf(0, 0, string(m, '0')); // minimize the sum of arr[i][j],
connected.

```

### 11.13 • Permutation DP

Let  $D_{i,j}$  be the number of pre-permutations that uses the value from 1 to  $i$ , and has  $j$  components. The transition is to:

- Create new component:  $(i, j) \rightarrow (i+1, j+1)$ .
- Append  $i+1$  to one of the component:  $(i, j) \rightarrow (i+1, j)$ .
- Connect two components using  $i+1$ :  $(i, j) \rightarrow (i+1, j-1)$ .

When the starting/ending point is specified, you need to be careful about the number of *open-ended* points.

### 11.14 • Berlekamp-Massey

Also check [Kitamasa](#) at page 14.

```

vector<ll> mulp(vector<ll> arr, vector<ll> brr){ // A \times B
    normp(arr); normp(brr);
    int al = arr.size(), bl = brr.size();
    vector<ll> res(al+bl-1);
    for (int i = 0; i < al; i++){
        for (int j = 0; j < bl; j++){
            res[i+j] += arr[i]*brr[j] % mod; res[i+j] %= mod;
        }
    } return res;
}
vector<ll> solve(vector<ll> arr){
    int n = arr.size();
    vector<ll> res, val; int pos = -1;
    for (int i = 0; i < n; i++){
        int rl = res.size();
        ll d = arr[i]; for (int j = 0; j < rl; j++){
            d -= res[j]*arr[i-j-1] % mod; d = (d+mod)%mod;
        } if (d == 0){ continue; }
        if (pos == -1){
            res.resize(i+1); for (int j = 0; j <= i; j++){ res[j] =
            1; }
            pos = i;
        } else{
            vector<ll> v{1}; for (ll x : val){
                v.push_back((mod-x)%mod); }
            int vl = v.size();
            ll c = 0; for (int j = 0; j < vl; j++){
                c += v[j]*arr[pos-j] % mod; c %= mod;
            } c = d*finv(c) % mod;
        }
    }
}

```

```

    for (ll& x : v){ x = x*c % mod; }
    vector<ll> tmp(i-pos-1, 0); for (ll x : v){
        tmp.push_back(x); }
    if (i - res.size() >= pos - val.size()){ val = res; pos
    = i; }
    int rl = max(res.size(), tmp.size()); res.resize(rl);
    for (int i = 0; i < rl; i++){ res[i] = (res[i]+tmp[i])
    % mod; }
    }
} return res;
}

```

## 12 String & Sequence

### 12.1 • Knuth-Morris-Pratt

```

int j = 0; for (int i = 1; i < m; i++){
    while (j != 0){
        if (t[i] == t[j]){ break; } else{ j = jmp[j-1]; }
    } if (t[i] == t[j]){ j += 1; jmp[i] = j; }
}
j = 0; for (int i = 0; i < n; i++){
    while (j != 0){
        if (s[i] == t[j]){ break; } else{ j = jmp[j-1]; }
    } if (s[i] == t[j]){
        if (j+1 == m){ /* matched on (i-m, i) */ j = jmp[j]; }
        else{ j += 1; }
    }
}

```

#### Compressed Value

```

int pos[130]; int cnt[130];
inline bool chk(const string& a, int as, int ae, const
string& b, int bs, int be){
    bool an = (pos[a[ae]] == -1), bn = (cnt[b[be]] == 0);
    if (an || bn){ return an == bn; }
    int p = pos[a[ae]]; int d = ae-p; // a[p] == a[ae].
    return (b[be-d] == b[be]);
}

inline void mvs(const string& s, int& st, int d){
    while (d--){ if (pos[s[st]] == st){ pos[s[st]] = -1; } st
    += 1; }
}

inline void mvi(const string& s, int& i, int d){
    while (d--){ pos[s[i]] = i; i += 1; }
}

inline void mvj(const string& s, int& j, int d){
    if (d == +1){ cnt[s[j]] += 1; j += 1; return; }
    d = -d; while (d--){ j -= 1; cnt[s[j]] -= 1; }
}

memset(pos, -1, sizeof(pos)); memset(cnt, 0, sizeof(cnt));
int st = 1, i = 1, j = 0; while (i < tl){
    while (j != 0){
        if (chk(t, st, i, t, 0, j)){ break; }
        else{ int p = jmp[j-1]; int d = j-p; mvs(t, st, +d);
        mvj(t, j, -d); }
    }
}

```

```

    } if (chk(t, st, i, t, 0, j)){ mvj(t, j, +1); jmp[i] = j;
    mvi(t, i, +1); }
    else{ mvi(t, i, +1); mvs(t, st, +1); }
}
memset(pos, -1, sizeof(pos)); memset(cnt, 0, sizeof(cnt));
st = 0; i = 0; j = 0; int ans = 0; while (i < sl){
    while (j != 0){
        if (chk(s, st, i, t, 0, j)){ break; }
        else{ int p = jmp[j-1]; int d = j-p; mvs(s, st, +d);
        mvj(t, j, -d); }
    } if (chk(s, st, i, t, 0, j)){
        if (j+1 == tl){
            ans += 1; int p = jmp[j]; int d = j-p; mvs(s, st, +d);
            mvj(t, j, -d); }
            mvi(s, i, +1); mvs(s, st, +1);
        }
        else{ mvj(t, j, +1); mvi(s, i, +1); }
    } else{
        int p = jmp[j-1]; int d = j-p; mvs(s, st, +d); mvj(t, j,
        -d);
        mvi(s, i, +1); mvs(s, st, +1);
    }
}
cout << ans;

```

### 12.2 • Z Algorithm

```

int l = 0, r = 0; for (int i = 1; i < sl; i++){
    if (i < r){ z[i] = min(r-i, z[i-l]); }
    while (i+z[i] < sl){
        if (s[z[i]] == s[i+z[i]]){ z[i] += 1; }
        else{ break; }
    } if (r < i+z[i]){ l = i; r = i+z[i]; }
} z[0] = sl;

```

### 12.3 • Aho-Corasick

```

int q; cin >> q; while (q--){
    string s; cin >> s; /* Trie */ psh(s);
}
queue<int> q; q.push(0); /* Root */ while (!q.empty()){
    int now = q.front(); q.pop();
    for (int nxi : nxtSet){
        int nxt = trie[now].nxt[nxi]; if (nxt == 0){ continue; }
        if (now == 0){ trie[nxt].jmp = now; }
        else{
            int ptr = trie[now].jmp; while (ptr != 0){
                if (trie[ptr].nxt[nxi] != 0){ break; } else{ ptr =
                trie[ptr].jmp; }
            } if (trie[ptr].nxt[nxi] != 0){ ptr =
            trie[ptr].nxt[nxi]; }
            trie[nxt].jmp = ptr;
        } q.push(nxt);
        trie[nxt].chk |= trie[trie[nxt].jmp].chk;
    }
}
int ptr = 0; for (int nxi : s){
    while (ptr != 0){
        if (trie[ptr].nxt[nxi] != 0){ break; } else{ ptr =
        trie[ptr].jmp; }
    }
}

```

```

    } if (trie[ptr].nxt[nxi] != 0){ ptr = trie[ptr].nxt[nxi]; }
    if (trie[ptr].chk){ ans += 1; }
}

```

### 12.4 • Suffix Array & Longest Common Prefix Array

```

int sa[1000020], pos[1000020], lcp[1000020];
int tmp[1000020], cnt[1000020], res[1000020];
void init(string s){
    s.push_back('#'); int sl = s.size();
    for (int i = 0; i < sl; i++){ sa[i] = i; }
    sort(sa, sa+sl, [&s](int s1, int s2){ return s[s1] < s[s2];
    });
    res[sa[0]] = 0; for (int i = 1; i < sl; i++){
        int s1 = sa[i-1], s2 = sa[i]; res[s2] = res[s1] + (s[s1]
        != s[s2]);
    }
    for (int k = 1; ; k *= 2){
        for (int i = 0; i < sl; i++){ cnt[i] = 0; tmp[i] =
        (sa[i]-k%sl+sl)%sl; }
        for (int i = 0; i < sl; i++){ cnt[res[i]] += 1; }
        for (int i = 1; i < sl; i++){ cnt[i] += cnt[i-1]; }
        for (int i = sl-1; i >= 0; i--){
            int sp = tmp[i]; int r = res[sp];
            int p = cnt[r]; sa[p-1] = sp; cnt[r] -= 1;
        }
        tmp[sa[0]] = 0; for (int i = 1; i < sl; i++){
            int s1 = sa[i-1], s2 = sa[i];
            int t1 = (s1+k)%sl, t2 = (s2+k)%sl;
            tmp[s2] = tmp[s1] + (res[s1] != res[s2] || res[t1] !=
            res[t2]);
        } memcpy(res, tmp, sizeof(res)); if (res[sa[sl-1]] ==
        sl-1){ break; }
    } s.pop_back(); sl -= 1;
    for (int i = 0; i < sl; i++){ sa[i] = sa[i+1]; }
    for (int i = 0; i < sl; i++){ pos[sa[i]] = i; }
    int l = 0; for (int i = 0; i < sl; i++){
        l = max(l-1, 0); int p1 = pos[i], p2 = pos[i]-1; if (p2 <
        0){ continue; }
        int s1 = sa[p1], s2 = sa[p2]; while (s1+1 < sl && s2+1 <
        sl){
            if (s[s1+1] != s[s2+1]){ break; } else{ l += 1; }
        } lcp[p2] = l;
    }
}

```

#### Traversing

```

int idx[1000020]; // idx[i] = NULL position of sa[i]. Useful
when S is multi-string.
// qry(st, ed) returns pair(min lcp[i], argmin) for [st, ed).
// solve on sa[st..ed][i..j).
ll solve(int st, int ed, int i){
    if (st == ed){ int j = idx[st]; return chk[st].fr <= ed &&
    ed < chk[st].sc ? j-i : 0; }
}

```



```

int j = qry(st, ed).fr; vector<int> mid{st};
while (mid.back() < ed){
    pi2 p = qry(mid.back(), ed); if (p.fr != j){ break; }
    mid.push_back(p.sc+1);
} mid.push_back(ed+1);
ll ans = (chk[st].fr <= ed && ed < chk[st].sc ? j-i : 0);
int len = mid.size(); for (int p = 1; p < len; p++){
    ans += solve(mid[p-1], mid[p]-1, j);
} return ans;
}

```

## 12.5 • Manacher's

```

string s = "#"; for (char c : inp){ s += c; s += '#'; }
int sl = s.size(); int ptr = -1, mid = -1;
for (int i = 0; i < sl; i++){
    if (ptr >= i){ pos[i] = min(ptr-i, pos[mid + mid-i]); }
    else{ pos[i] = 0; }
    int i1 = i-pos[i], i2 = i+pos[i];
    while (0 <= i1-1 && i2+1 < sl){
        if (s[i1-1] == s[i2+1]){ i1--; i2++; pos[i] += 1; }
        else{ break; }
    } if (i2 > ptr){ ptr = i2; mid = i; }
} // length = *max_element(pos, pos+sl);

```

## 12.6 • Rolling Hash

```

const int H = 2;
typedef array<ll, H> alH;
const ll mod[H] = {993244853, 998244853};
const ll mul[H] = {31, 37};
ll ppw[H][N+20], prf[H][N+20];
void init(int hi, const string& s){ int sl = s.size();
    ppw[hi][0] = 1; for (int i = 1; i <= N; i++){ ppw[hi][i] =
        ppw[hi][i-1]*mul[hi] % mod[hi]; }
    for (int i = 1; i <= sl; i++){ prf[hi][i] =
        (prf[hi][i-1]*mul[hi] + s[i]) % mod[hi]; }
}
inline alH hsh(int st, int ed){
    alH res; for (int hi = 0; hi < H; hi++){
        res[hi] = (prf[hi][ed] - prf[hi][st-1]*ppw[hi][ed-st+1] %
            mod[hi] + mod[hi]) % mod[hi];
    } return res;
}

```

## 12.7 • Iterating Every Permutation

```

int arr[12]; pi2 pos[12]; // Init = 0

for (int i = 1; i <= n; i++){ arr[i] = i; pos[i] = {i, (i==1
? 0 : -1)}; }
arr[0] = arr[n+1] = n+1; while (1){
    for (int i = 1; i <= n; i++){ cout << arr[i] << ' '; } cout
<< endl;
    int x = n; while (x >= 1){
        if (pos[x].sc == 0){ x -= 1; } else{ break; }
    } if (x == 0){ break; }
    int xp = pos[x].fr, xd = pos[x].sc;
    int yp = xp+xd; int y = arr[yp];

```

```

pos[x].fr += xd; pos[y].fr -= xd; swap(arr[xp], arr[yp]);
if (arr[yp+xd] > x){ pos[x].sc = 0; }
for (int y = x+1; y <= n; y++){
    if (pos[y].sc != 0){ continue; }
    int yp = pos[y].fr; pos[y].sc = (xp < yp ? -1 : +1);
}
}

```

## 13 Data Structure

### 13.1 • Li-Chao Tree

Find  $\max f_i(x) = a_i x + b_i$ , with line insertion update.

- Time Complexity:  $\mathcal{O}(\log N)$  per update/query..

```

// Max version. if Min then use the commented version.
inline ll f(pl2 l, ll x){ return l.fr*x + l.sc; }
struct Node{ pl2 l = {0, -INF /* INF */}; pi2 nxt = {0, 0};
};
vector<Node> seg;
void upd(int ni, ll ns, ll ne, pl2 l1){
    pl2 l2 = seg[ni].l;
    if (f(l1, ns) < /* > */ f(l2, ns)){ swap(l1, l2); }
    if (f(l1, ne) >= /* <= */ f(l2, ne)){ seg[ni].l = l1;
        return; }
    ll nm = ns+ne >> 1;
    if (f(l1, nm) >= /* <= */ f(l2, nm)){
        seg[ni].l = l1; if (seg[ni].nxt.sc == 0){
            seg[ni].nxt.sc = seg.size(); seg.emplace_back();
        } upd(seg[ni].nxt.sc, nm+1, ne, l2);
    }
    else{
        seg[ni].l = l2; if (seg[ni].nxt.fr == 0){
            seg[ni].nxt.fr = seg.size(); seg.emplace_back();
        } upd(seg[ni].nxt.fr, ns, nm, l1);
    }
}
inline void upd(pl2 l){ return upd(0, -X, X, l); }
ll qry(int ni, ll ns, ll ne, ll x){
    pl2 l = seg[ni].l; ll nm = ns+ne >> 1;
    int nxt = (x <= nm ? seg[ni].nxt.fr : seg[ni].nxt.sc);
    if (nxt == 0){ return f(l, x); }
    else{
        if (x <= nm){ return max /* min */(f(l, x), qry(nxt, ns,
            nm, x)); }
        else{ return max /* min */(f(l, x), qry(nxt, nm+1, ne,
            x)); }
    }
}
} inline ll qry(ll x){ return qry(0, -X, X, x); }

```

### 13.2 • Multi-Dimensional Segment Tree

```

const int N = 1024;
int seg[2050][2050];
void updx(int yp, int x, int val){ int xp = x+N-1;
    if (yp >= N){ seg[yp][xp] = val; }
    else{ seg[yp][xp] = max(seg[yp<<1][xp], seg[yp<<1|1][xp]);
    }
    xp >>= 1; while (xp){ seg[yp][xp] = max(seg[yp][xp<<1],
        seg[yp][xp<<1|1]); xp >>= 1; }
}

```

```

}
void upd(int y, int x, int val){ int yp = y+N-1;
    updx(yp, x, val); yp >>= 1;
    while (yp){ updx(yp, x, val); yp >>= 1; }
}
int qryx(int yp, int x1, int x2){ x1 += N-1; x2 += N-1;
    int res = 0; while (x1 <= x2){
        if (x1 & 1){ res = max(seg[yp][x1], res); x1 += 1; }
        if (~x2 & 1){ res = max(seg[yp][x2], res); x2 -= 1; }
        if (x1 > x2){ break; } x1 >>= 1; x2 >>= 1;
    } return res;
}
}
int qry(int y1, int y2, int x1, int x2){ y1 += N-1; y2 +=
N-1;
    int res = 0; while (y1 <= y2){
        if (y1 & 1){ res = max(qryx(y1, x1, x2), res); y1 += 1; }
        if (~y2 & 1){ res = max(qryx(y2, x1, x2), res); y2 -= 1;
        }
        if (y1 > y2){ break; } y1 >>= 1; y2 >>= 1;
    } return res;
}
}

```

### 13.3 • Persistent Segment Tree

You can use this to make Persistent Array. You can then use this Persistent Array to make Persistent Queue. Persistent Stack is just Trie.

```

const int N = 1000000;
struct Node{ int sum; int nxt[2]; };
vector<Node> seg; vector<int> root;

void upd(int ni, int ns, int ne, int qi, int qx){
    if (ns == ne){ seg[ni].sum += qx; return; }
    int nm = ns+ne >> 1; int nxi = (qi <= nm ? 0 : 1);
    int p = seg[ni].nxt[nxi]; seg[ni].nxt[nxi] = seg.size();
    seg.push_back(seg[p]);
    if (qi <= nm){ upd(seg[ni].nxt[nxi], ns, nm, qi, qx); }
    else{ upd(seg[ni].nxt[nxi], nm+1, ne, qi, qx); }
    seg[ni].sum += qx;
}
}
inline void upd(int idx, int pos, int val){
    root.push_back(seg.size()); seg.push_back(seg[root[idx]]);
    upd(root.back(), 0, N, pos, val);
}
}
int qry(int ni, int ns, int ne, int qs, int qe){
    if (qs <= ns && ne <= qe){ return seg[ni].sum; }
    int nm = ns+ne >> 1;
    int res = 0;
    if (qs <= nm && seg[ni].nxt[0]){ res += qry(seg[ni].nxt[0],
        ns, nm, qs, qe); }
    if (nm+1 <= qe && seg[ni].nxt[1]){ res +=
        qry(seg[ni].nxt[1], nm+1, ne, qs, qe); }
    return res;
}
}
inline int qry(int idx, int st, int ed){
    return qry(root[idx], 0, N, st, ed);
}
}

```

### 13.4 • Segment Tree Beats

Lazy Segtree, but less lazy and more planning.

- Time Complexity:  $\mathcal{O}(Q \log N)$ , amortized..

```
struct Node{ int mx1, mx2, mxc; int sum; int laz; }; //
update on a = min(a, x)
inline Node mrg(const Node& l, const Node& r){ Node p;
    p.mx1 = max(l.mx1, r.mx1);
    p.mx2 = max((l.mx1 == p.mx1 ? l.mx2 : l.mx1), (r.mx1 ==
    p.mx1 ? r.mx2 : r.mx1));
    p.mxc = (l.mx1 == p.mx1 ? l.mxc : 0) + (r.mx1 == p.mx1 ?
    r.mxc : 0);
    p.sum = l.sum + r.sum;
    p.laz = INF; return p;
}
Node seg[2097152];
inline void pro(int ni, int ns, int ne){
    if (ns != ne){
        seg[ni<<1].laz = min(seg[ni<<1].laz, seg[ni].laz);
        seg[ni<<1|1].laz = min(seg[ni<<1|1].laz, seg[ni].laz);
    }
    ll dif = seg[ni].laz - seg[ni].mx1; if (dif <= 0){
        seg[ni].sum += dif * seg[ni].mxc;
        seg[ni].mx1 += dif;
    } seg[ni].laz = INF;
}
void upd(int ni, int ns, int ne, int qs, int qe, int qx){
    pro(ni, ns, ne);
    if (qe < ns || ne < qs || seg[ni].mx1 <= qx){ return; }
    if (qs <= ns && ne <= qe && seg[ni].mx2 < qx){ seg[ni].laz
    = qx; return pro(ni, ns, ne); }
    int nm = ns+ne >> 1;
    upd(ni<<1, ns, nm, qs, qe, qx); upd(ni<<1|1, nm+1, ne, qs,
    qe, qx);
    seg[ni] = mrg(seg[ni<<1], seg[ni<<1|1]);
} inline void upd(int st, int ed, int val){ return upd(1, 1,
N, st, ed, val); }
Node qry(int ni, int ns, int ne, int qs, int qe){
    pro(ni, ns, ne);
    if (qe < ns || ne < qs){ return {-1, -1, 0, 0, INF}; }
    if (qs <= ns && ne <= qe){ return seg[ni]; }
    int nm = ns+ne >> 1;
    return mrg(qry(ni<<1, ns, nm, qs, qe), qry(ni<<1|1, nm+1,
    ne, qs, qe));
} inline Node qry(int st, int ed){ return qry(1, 1, N, st,
ed); }
```

### 13.5 • Gomory-Hu Tree

```
int gph[220][220];
const int INF = 1e9;
int flow(int st, int ed){
    memcpy(adj, gph, sizeof(adj)); adj[src][st] = INF;
    adj[ed][snk] = INF;
    /* Find MinCut between st and ed and return the value */
}
```

```
int par[220]; int val[220];
vector<int> tree[220]; pi3 edg[220];
```

```
int n, m; cin >> n >> m; src = 0; snk = n+1; nc = n+2;
while (m--){
    int v, w, x; cin >> v >> w >> x;
    gph[v][w] += x; gph[w][v] += x;
}
for (int v = 2; v <= n; v++){ par[v] = 1; }
for (int v = 2; v <= n; v++){
    int p = par[v]; int res = flow(v, p);
    val[v] = res;
    for (int i = 1; i <= n; i++){
        if (i == v){ continue; }
        if (par[i] != p){ continue; }
        if (dis[i] != -1){ par[i] = v; }
    }
    if (dis[par[p]] != -1){
        par[v] = par[p]; par[p] = v;
        val[v] = val[p]; val[p] = res;
    }
}
int cnt = 0; for (int v = 1; v <= n; v++){
    int p = par[v], x = val[v]; if (p == 0){ continue; }
    edg[++cnt] = {{v, p}, x}; tree[v].push_back(cnt);
    tree[p].push_back(cnt);
}
```

### 13.6 • Centroid Tree

Repeatedly apply centroid to the subtrees, and make the tree. The height of the centroid tree is at most  $\lfloor \log_2 N \rfloor + 1$ .

Also, the path  $v \Rightarrow w$  from original tree always pass through LCA of  $v$  and  $w$  in centroid tree.

```
vector<int> tree[100020]; int par[100020];
int dfs(int v){
    centroid(v); v = cent.fr; chk[v] = 1;
    for (pi2 e : adj[v]){
        int w = e.fr; if (chk[w]){ continue; }
        int p = dfs(w); tree[v].push_back(p); par[p] = v;
    } return v;
}
```

### 13.7 • Splay Tree

$k$  is 1-indexed, assuming dummy on both end. insert inserts after  $k$ -th vertex (front if  $k = 0$ ).

- Time Complexity:  $\mathcal{O}(Q \log N)$ , amortized..

```
const int INF = 1e9;
struct Node{
    int p = -1, l = -1, r = -1; bool flp = 0; int val = 0;
    int cnt = 1; ll sum = 0; int mn = INF, mx = -INF;
}; vector<Node> tree; int root = -1;
Node init(int x){
    Node p; p.cnt = 1; p.val = x;
    p.sum = p.mn = p.mx = x; return p;
} void put(int ni, const Node p){
```

```
tree[ni].cnt = p.cnt; tree[ni].sum = p.sum; tree[ni].mn =
p.mn; tree[ni].mx = p.mx;
}
Node mrg(int li, int ri){
    const Node &l = tree[li], &r = tree[ri];
    Node p; p.cnt = l.cnt + r.cnt;
    p.sum = l.sum + r.sum; p.mn = min(l.mn, r.mn); p.mx =
    max(l.mx, r.mx);
    return p;
}
void propagate(int ni){
    if (ni == -1){ return; } if (tree[ni].flp == 0){ return; }
    swap(tree[ni].l, tree[ni].r);
    if (tree[ni].l != -1){ tree[tree[ni].l].flp ^= 1; }
    if (tree[ni].r != -1){ tree[tree[ni].r].flp ^= 1; }
    tree[ni].flp = 0;
}
void update(int ni){
    put(ni, init(tree[ni].val));
    if (tree[ni].l != -1){ propagate(tree[ni].l); put(ni,
    mrg(tree[ni].l, ni)); }
    if (tree[ni].r != -1){ propagate(tree[ni].r); put(ni,
    mrg(ni, tree[ni].r)); }
} inline void update(int ni, ll val){ tree[ni].val = val;
return update(ni); }
void rotate(int ni){
    propagate(tree[ni].p); propagate(ni);
    int p1 = tree[ni].p; if (p1 == -1){ return; }
    int p2 = tree[p1].p; if (p2 != -1){
        (tree[p2].l == p1 ? tree[p2].l : tree[p2].r) = ni;
        tree[ni].p = p2;
    } else{ root = ni; tree[ni].p = -1; }
    int p3 = -1; if (tree[p1].l == ni){
        tree[p1].l = p3 = tree[ni].r; tree[ni].r = p1;
    } else{
        tree[p1].r = p3 = tree[ni].l; tree[ni].l = p1;
    } tree[p1].p = ni; if (p3 != -1){ tree[p3].p = p1; }
    update(p1); update(ni);
}
int splay(int ni, int top = -1){
    while (tree[ni].p != top){
        propagate(tree[tree[ni].p].p); propagate(tree[ni].p);
        propagate(ni);
        int p1 = tree[ni].p; int p2 = tree[p1].p;
        if (p2 != top){ rotate((tree[p2].l==p1) ==
        (tree[p1].l==ni) ? p1 : ni); }
        rotate(ni);
    } propagate(ni); return ni;
}
int kth(int k){
    int ni = root; while (1){
        propagate(ni);
        int cnt = (tree[ni].l == -1 ? 0 : tree[tree[ni].l].cnt);
        if (cnt == k){ break; }
        if (k < cnt){ ni = tree[ni].l; }
        else{ k -= cnt+1; ni = tree[ni].r; }
    } return splay(ni);
}
```

```

}
int segment(int st, int ed){
    kth(ed+1); int r = root; kth(st-1); splay(r, root); return
    tree[tree[root].r].l;
}
void insert(int k, int val){
    if (root == -1){ root = tree.size();
    tree.push_back(init(val)); return; }
    int ni = root; while (1){
        int cnt = (tree[ni].l == -1 ? 0 : tree[tree[ni].l].cnt);
        if (k < cnt){
            if (tree[ni].l == -1){
                tree[ni].l = tree.size(); tree.push_back(init(val));
                tree[tree[ni].l].p = ni; ni = tree[ni].l; break;
            } else{ ni = tree[ni].l; }
        } else{
            if (tree[ni].r == -1){
                tree[ni].r = tree.size(); tree.push_back(init(val));
                tree[tree[ni].r].p = ni; ni = tree[ni].r; break;
            } else{ k -= cnt+1; ni = tree[ni].r; }
        }
    } splay(ni);
}
void erase(int k){
    segment(k, k); tree[tree[root].r].l = -1;
    splay(tree[root].r);
}
}
void rev(int st, int ed){
    if (st > ed){ return; }
    int ni = segment(st, ed); tree[ni].flp ^= 1; propagate(ni);
}

```

### 13.8 • Link-Cut Tree

Also check [Splay Tree](#) at page 18.

```

bool isroot(int ni){
    if (tree[ni].p == -1){ return 1; }
    return tree[tree[ni].p].l != ni && tree[tree[ni].p].r !=
    ni;
}
void rotate(int ni){
    propagate(tree[ni].p); propagate(ni);
    int p1 = tree[ni].p; if (isroot(ni)){ return; }
    int p2 = tree[p1].p; if (!isroot(p1)){
        (tree[p2].l == p1 ? tree[p2].l : tree[p2].r) = ni;
        tree[ni].p = p2;
    } else{ tree[ni].p = p2; }
    int p3 = -1; if (tree[p1].l == ni){
        tree[p1].l = p3 = tree[ni].r; tree[ni].r = p1;
    } else{
        tree[p1].r = p3 = tree[ni].l; tree[ni].l = p1;
    } tree[p1].p = ni; if (p3 != -1){ tree[p3].p = p1; }
    update(p1); update(ni);
}
int splay(int ni, int top = -1){
    while (!isroot(ni)){

```

```

        propagate(tree[tree[ni].p].p); propagate(tree[ni].p);
        propagate(ni);
        int p1 = tree[ni].p; int p2 = tree[p1].p;
        if (!isroot(p1)){ rotate((tree[p2].l==p1) ==
        (tree[p1].l==ni) ? p1 : ni); }
        rotate(ni);
    } propagate(ni); return ni;
}
void access(int ni){
    splay(ni); tree[ni].r = -1;
    while (tree[ni].p != -1){
        splay(tree[ni].p); tree[tree[ni].p].r = ni; splay(ni);
    }
}
void reroot(int ni){
    access(ni); splay(ni); tree[ni].flp ^= 1;
}
int parent(int ni){
    access(ni); propagate(ni);
    if (tree[ni].l == -1){ return -1; }
    ni = tree[ni].l; propagate(ni);
    while (tree[ni].r != -1){ ni = tree[ni].r; propagate(ni); }
    return splay(ni);
}
int find(int ni){
    access(ni); propagate(ni);
    if (0){
        int n = tree.size();
        for (int i = 0; i < n; i++){
            cout << "Node " << i << ": " << tree[i].p << ", " <<
            tree[i].l << " " << tree[i].r << endl;
            cout << tree[i].flp << " " << tree[i].val << " " <<
            tree[i].sum << endl;
        } cout << endl << flush;
    }
    while (tree[ni].l != -1){ ni = tree[ni].l; propagate(ni); }
    return splay(ni);
}
void link(int ni, int pi){
    reroot(ni); access(ni); access(pi); tree[ni].l = pi;
    tree[pi].p = ni; update(ni);
}
void cut(int ni){
    access(ni); tree[tree[ni].l].p = -1; tree[ni].l = -1;
    update(ni);
}
int lca(int vi, int wi){
    access(vi); access(wi); splay(vi);
    if (tree[vi].p != -1){ return tree[vi].p; } else{ return
    vi; }
}
// Connected? find(vi) == find(wi)
void upd(int vi, ll x){
    splay(vi); update(vi, x);
}
void upd(int vi, int wi, ll x){

```

```

    int r = find(vi);
    reroot(vi); access(wi); splay(vi); tree[vi].laz += x;
    propagate(vi);
    reroot(r);
    // Edge Update: v - l - w excluding l.
    int li = lca(vi, wi);
    access(li); splay(li); propagate(li); update(li,
    tree[li].val-x);
}
ll qry(int vi, int wi){
    int li = lca(vi, wi); ll res = tree[li].val;
    access(vi); splay(li); if (tree[li].r != -1){ res +=
    tree[tree[li].r].sum; }
    access(wi); splay(li); if (tree[li].r != -1){ res +=
    tree[tree[li].r].sum; }
    return res;
}

```

### 13.9 • Permutation Tree

upd and qry is range add update and range min query, using Lazy-Seg.

```

for (int i = 1; i <= n; i++){ val[i] = {arr[i], arr[i]};
pos[i] = {i, i}; }
vector<int> mxs{0}, mns{0};
int ni = n+1; vector<int> stk; for (int i = 1; i <= n; i++){
    while (mxs.size() > 1){
        int sl = mxs.size(); int j = mxs[sl-1];
        if (arr[j] > arr[i]){ break; }
        upd(mxs[sl-2]+1, mxs[sl-1], arr[i]-arr[j]);
        mxs.pop_back();
    } mxs.push_back(i);
    while (mns.size() > 1){
        int sl = mns.size(); int j = mns[sl-1];
        if (arr[j] < arr[i]){ break; }
        upd(mns[sl-2]+1, mns[sl-1], arr[j]-arr[i]);
        mns.pop_back();
    } mns.push_back(i);
    stk.push_back(i); while (stk.size() > 1){
        int sl = stk.size();
        int a = stk[sl-2]; int b = stk[sl-1];
        if (val[a].sc+1 == val[b].fr){
            if (dir[a] == +1){
                val[a].sc = val[b].sc; pos[a].sc = pos[b].sc;
                adj[a].push_back(b); stk.pop_back();
            } else{
                val[ni] = {val[a].fr, val[b].sc}; pos[ni] =
                {pos[a].fr, pos[b].sc};
                adj[ni].push_back(a); adj[ni].push_back(b); dir[ni] =
                +1;
                stk.pop_back(); stk.pop_back(); stk.push_back(ni++);
            }
        } else if (val[b].sc+1 == val[a].fr){
            if (dir[a] == -1){
                val[a].fr = val[b].fr; pos[a].sc = pos[b].sc;
                adj[a].push_back(b); stk.pop_back();
            }
        }
    }
}

```

```

    } else{
        val[ni] = {val[b].fr, val[a].sc}; pos[ni] =
        {pos[a].fr, pos[b].sc};
        adj[ni].push_back(a); adj[ni].push_back(b); dir[ni] =
        -1;
        stk.pop_back(); stk.pop_back(); stk.push_back(ni++);
    }
} else{
    if (qry(1, pos[a].sc) != 0){ break; }
    val[ni] = {min(val[a].fr, val[b].fr), max(val[a].sc,
    val[b].sc)};
    pos[ni] = {min(pos[a].fr, pos[b].fr), max(pos[a].sc,
    pos[b].sc)};
    adj[ni].push_back(b); adj[ni].push_back(a); dir[ni] =
    0;
    stk.pop_back(); stk.pop_back(); while (!stk.empty()){
        if (val[ni].sc-val[ni].fr == pos[ni].sc-pos[ni].fr){
            break; }
        int p = stk.back(); stk.pop_back();
        val[ni].fr = min(val[ni].fr, val[p].fr);
        val[ni].sc = max(val[ni].sc, val[p].sc);
        pos[ni].fr = min(pos[ni].fr, pos[p].fr);
        pos[ni].sc = max(pos[ni].sc, pos[p].sc);
        adj[ni].push_back(p);
    } reverse(adj[ni].begin(), adj[ni].end());
    stk.push_back(ni++);
}
} upd(1, i, -1);
} int r = stk.front();

```

## 14 Query & Decomposition

### 14.1 Fractional Binary Search

```

const ll X = 1e9;
p12 l = {0, 1}, r = {1, 0}; // assume f(l) = false and f(r) =
true
bool chk = 1; while (chk){
    chk = 0;
    ll res = 0; bool flg = 1; for (ll bit = 1; bit; flg ?
    (bit<=1) : (bit>=1)){
        p12 mid = {l.fr + r.fr*(res+bit), l.sc +
        r.sc*(res+bit)};
        if (mid.fr > X || mid.sc > X){ flg = 0; }
        else{ if (!f(mid)){ res += bit; } else{ flg = 0; } }
    } l = {l.fr + r.fr*res, l.sc + r.sc*res}; chk |= (res >
    0);
    res = 0; flg = 1; for (ll bit = 1; bit; flg ? (bit<=1) :
    (bit>=1)){
        p12 mid = {l.fr*(res+bit) + r.fr, l.sc*(res+bit) +
        r.sc};
        if (mid.fr > X || mid.sc > X){ flg = 0; }
        else{ if (f(mid)){ res += bit; } else{ flg = 0; } }
    } r = {l.fr*res + r.fr, l.sc*res + r.sc}; chk |= (res >
    0);
}

```

### 14.2 Heavy-Light Decomposition

- Time Complexity:  $\mathcal{O}(T(N) \log N)$  per update/query..

```

vector<int> adj[500020]; int par[500020];
int siz[500020], dep[500020];
void dfs1(int now, int pre){
    siz[now] = 1; par[now] = pre; erase(adj[now], pre);
    for (int& nxt : adj[now]){
        dep[nxt] = dep[now]+1; dfs1(nxt, now);
        siz[now] += siz[nxt];
        if (siz[nxt] >= siz[adj[now][0]]){ swap(nxt,
        adj[now][0]); }
    }
}
pi2 ord[500020]; int ont;
int chn[500020];
void dfs2(int now, int pre){
    ord[now].fr = ++ont;
    for (int nxt : adj[now]){
        if (nxt == adj[now][0]){ chn[nxt] = chn[now]; }
        else{ chn[nxt] = nxt; } dfs2(nxt, now);
    }
    ord[now].sc = ont;
}
// dfs1(root, root); chn[root] = root; dfs2(root, root);
void upd_pth(int v, int w, int x){
    while (chn[v] != chn[w]){
        if (dep[chn[v]] > dep[chn[w]]){ swap(v, w); }
        upd(ord[chn[w]].fr, ord[w].fr, x); w = par[chn[w]];
    }
    if (dep[v] > dep[w]){ swap(v, w); } upd(ord[v].fr,
    ord[w].fr, x);
    // if Edge Weight: if v == w then no update. otherwise v =
    chl[v][0] before update
}
u32 qry_pth(int v, int w){
    u32 res = 0; while (chn[v] != chn[w]){
        if (dep[chn[v]] > dep[chn[w]]){ swap(v, w); }
        res += qry(ord[chn[w]].fr, ord[w].fr); w = par[chn[w]];
    }
    if (dep[v] > dep[w]){ swap(v, w); }
    return res+qry(ord[v].fr, ord[w].fr);
    // if Edge Weight: if v == w then no query. otherwise v =
    chl[v][0] before query
}

```

### 14.3 Centroid Decomposition

- Time Complexity:  $\mathcal{O}(N \log N)$ ..

```

bool chk[100020]; int siz[100020];
void sizf(int v, int p){
    siz[v] = 1; for (pi2 e : adj[v]){
        int w = e.fr; if (chk[w]){ continue; } if (w == p){
        continue; }
        sizf(w, v); siz[v] += siz[w];
    }
}

```

```

pi2 cent = {-1, -1}; void centroid(int v, int p, int n){
    bool flg = 1; for (pi2 e : adj[v]){
        int w = e.fr; if (chk[w]){ continue; } if (w == p){
        continue; }
        centroid(w, v, n); if (siz[w]*2 > n){ flg = 0; }
    } if (flg && (n-siz[v])*2 <= n){ (cent.fr == -1 ? cent.fr :
    cent.sc) = v; }
} inline void centroid(int v){ cent = {-1, -1}; sizf(v, v);
centroid(v, v, siz[v]); }

```

```

ll dnc(int v){
    centroid(v); v = cent.fr;
    // solve on path that passes v. divide it by subtree, and
    merge subtree-wise. use top-down.
    chk[v] = 1; for (pi2 e : adj[v]){
        int w = e.fr; if (chk[w]){ continue; }
        ans += dnc(w);
    } return ans;
}

```

### 14.4 Tree Compression

- Time Complexity:  $\mathcal{O}(K \log N)$  per query..

```

vector<int> arr;
for (int i = 1; i <= n; i++){ int v; cin >> v;
arr.push_back(v); }
sort(arr.begin(), arr.end(), [](int v, int w){ return ord[v]
< ord[w]; });
for (int i = 1; i < n; i++){
    int v = arr[i-1], w = arr[i]; arr.push_back(lca(v, w));
} sort(arr.begin(), arr.end(), [](int v, int w){ return
ord[v] < ord[w]; });
arr.erase(unique(arr.begin(), arr.end()), arr.end());
int al = arr.size(); for (int i = 1; i < al; i++){
    ll l = lca(arr[i-1], arr[i]);
    adj[cvt(arr, l)].push_back({i, dis[arr[i]]-dis[l]});
    adj[i].push_back({cvt(arr, l), dis[arr[i]]-dis[l]});
}

```

### 14.5 Parallel Binary Search

- Time Complexity:  $\mathcal{O}(Q \log X)$ ..

```

// Binary Search: path available after ans-th event.
// 0-th event: impossible as nothing is active
// m-th event: possible as everything is connected
struct Query{ int st, ed; int idx; int v, w; };
Query qrr[100020];
while (1){
    sort(qrr+1, qrr+q+1, [](Query& q1, Query& q2){
        return (q1.st+q1.ed)/2 < (q2.st+q2.ed)/2;
    });
    for (int i = 1; i <= n; i++){ par[i] = -1; }
    bool flg = 0;
    int qi = 1; for (int ei = 0; ei <= n; ei++){
        if (ei > 0){ // Update
            int v = evt[ei].fr;
            par[v] = v; for (pi2 p : adj[v]){

```



```

    int w = p.fr; if (par[w] != -1){ uni(v, w); }
}
}
while (qi <= q){ // Try answering the queries
    if (qrr[qi].st+1 > qrr[qi].ed-1){ qi++; continue; }
    int mid = qrr[qi].st+qrr[qi].ed >> 1; flg = 1;
    if (mid != ei){ break; }
    int v = qrr[qi].v, w = qrr[qi].w;
    if (par[v] != -1 && par[w] != -1 && fnd(v) == fnd(w)){
        qrr[qi].ed = mid; }
    else{ qrr[qi].st = mid; } qi++;
}
} if (!flg){ break; }
} sort(qrr+1, qrr+q+1, [](Query& q1, Query& q2){ return
q1.idx < q2.idx; });
for (int i = 1; i <= q; i++){ cout << evt[qrr[i].ed].sc <<
endl; } // ed = first possible

```

## 14.6 • CDQ Divide and Conquer

3D LIS. Find longest chain  $(A_i, B_i, C_i)$  s.t.  $A_i < A_j$ ;  $B_i < B_j$ ;  $C_i < C_j$  holds.

```

// upd(p, x, t) / t = 0: A[p] = x / t = 1: A[p] = max(A[p],
x)
// qry(s, e) / max A[s..e]
// Assume arr is sorted via A.
void dnc(int st, int ed){
    if (st == ed){ arr[st].dp = max(arr[st].dp, 1); return; }
    int mid = st+ed >> 1; dnc(st, mid);
    sort(arr+st, arr+mid+1, [](const T& a, const T& b){ return
a.b < b.b; });
    vector<int> qrr; for (int i = mid+1; i <= ed; i++){
        qrr.push_back(i); }
    sort(qrr.begin(), qrr.end(), [](int i1, int i2){ return
arr[i1].b < arr[i2].b; });
    int ptr = st; for (int i : qrr){
        while (ptr <= mid){
            if (arr[ptr].b < arr[i].b){ upd(arr[ptr].c,
arr[ptr].dp); ptr += 1; }
            else{ break; }
        } arr[i].dp = max(arr[i].dp, qry(1, arr[i].c)+1);
    } for (int i = st; i <= mid; i++){ upd(arr[i].c, 0); }
    return dnc(mid+1, ed);
}

```

## 15 Greedy

### 15.1 • Job Scheduling w/ Deadline & Duration

```

p12 arr[250020]; // {deadline, duration}
sort(arr+1, arr+n+1, [](p12 p1, p12 p2){ return p1.fr >
p2.fr; });
priority_stack<ll> pq; int ans = 0;
int i = 1; while (i <= n){
    int j = i; while (j <= n){
        if (arr[i].fr == arr[j].fr){ j += 1; } else{ break; }
    }
    for (int p = i; p < j; p++){ pq.push(arr[p].sc); }
}

```

```

ll tim = arr[i].fr - arr[j].fr; while (tim > 0){
    if (pq.empty()){ break; } ll t = pq.top(); pq.pop();
    ll d = min(t, tim); if (t-d == 0){ tim -= d; ans += 1; }
    else{ tim -= d; t -= d; pq.push(t); }
} i = j;
} cout << ans;

```

### 15.2 • Bounded Scheduling

If  $x \geq B_i$ , then  $x \leftarrow x + A_i$ .  $B_i + A_i \geq 0$ .

```

p12 arr[100020]; // (A[i], B[i])
sort(arr+1, arr+n+1, [](p12 p1, p12 p2){
    ll a1 = p1.fr, b1 = p1.sc;
    ll a2 = p2.fr, b2 = p2.sc;
    if ((a1 >= 0) != (a2 >= 0)){ return (a1 >= 0) > (a2 >= 0); }
    if (a1 >= 0){ return b1 < b2; } else{ return a1+b1 > a2+b2; }
});
ll x = 0; for (int i = 1; i <= n; i++){
    if (x < arr[i].sc){ /* Impossible; Quit */ }
    x += arr[i].fr;
} /* Possible */

```

## 16 Heuristics

Let the candidate be  $S$ , and  $f$  be the scoring function. Lower score is better. Hyperparameter can be changed based on the problem.

It is strictly recommended to return the global minimum, rather than the final result. If possible, alter the score rather than completely recalculate it.

### 16.1 • Simulated Annealing

Hyperparameter:  $d = 0.9999$  (higher = faster cooling),  $k = 10$  (higher = more volatile)

1. Set  $t = 1$  and generate random answer  $S$ , and compute the score  $F = f(S)$ .
2. Repeat the following:
  - (a) Randomly change the answer  $S$  to generate  $S'$ .
  - (b) Compute  $p = e^{(f(S)-f(S'))/kt}$ .
  - (c) Change the answer to  $S'$  with probability  $\min(p, 1)$ .

### 16.2 • Diversified Late Acceptance Search

Hyperparameter:  $l = 5$  (higher = longer memory)

1. Generate random answer  $S$ , and compute the score  $F = f(S)$ .
2. Set  $P_i = F$  for all  $1 \leq i \leq l$ . Set  $k = 0$ .
3. Repeat the following:
  - (a) Set  $F_* = F$  and generate  $S'$ , and compute  $F' = f(S')$ .
  - (b) If  $F' = F$  or  $F' < \max P$ , change the answer (and score) to  $S'$  (and  $F'$ ).
  - (c) If  $F > P_k$  or  $(F < P_i$  and  $F < F_*)$ , set  $P_i = F$ .
  - (d) Set  $k = (k + 1) \bmod l$ .

## 17 Miscellaneous

### 17.1 • List of Primes

- $\leq 10^3$ : 168 primes, max 997.
- $\leq 10^5$ : 1 229 primes, max 99 991.
- $\leq 10^6$ : 9 592 primes, max 999 983.
- $\leq 10^9$ : 50 847 534 primes, max 999 999 937.
- $\leq 10^{12}$ : max 999 999 999 989.
- $\leq 10^{18}$ : max 999 999 999 999 999 989.
- $998\,244\,353 = 119 \times 2^{23} + 1$  with  $\omega = 3$ .
- $167\,772\,161 = 5 \times 2^{26} + 1$  with  $\omega = 3$ .
- $469\,762\,049 = 13 \times 2^{27} + 1$  with  $\omega = 3$ .
- $2\,281\,701\,377 = 17 \times 2^{27} + 1$  with  $\omega = 3$ .
- $2\,483\,027\,969 = 37 \times 2^{26} + 1$  with  $\omega = 3$ .
- $2\,013\,265\,921 = 15 \times 2^{27} + 1$  with  $\omega = 31$ .

### 17.2 • List of Highly Composite Numbers

- 840: 32 divisors (3, 1, 1, 1)
- 83 160: 128 divisors (3, 3, 1, 1, 1)
- 720 720: 240 divisors (4, 2, 1, 1, 1, 1)
- 735 134 400: 1 344 divisors (6, 3, 2, 1, 1, 1, 1)
- 963 761 198 400: 6 720 divisors (6, 4, 2, 1, 1, 1, 1, 1)
- 897 612 484 786 617 600: 103 680 divisors (8, 4, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1)

### 17.3 • ASCII Table

|   | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7   |
|---|-----|-----|-----|-----|-----|-----|-----|-----|
|   | 8   | 9   | A   | B   | C   | D   | E   | F   |
| 0 | NUL | SOH | STX | ETX | EOT | ENQ | ACK | BEL |
|   | BS  | TAB | LF  | VT  | FF  | CR  | SO  | SI  |
| 1 | DLE | DC1 | DC2 | DC3 | DC4 | NAK | SYN | ETB |
|   | CAN | EM  | SUB | ESC | FS  | GS  | RS  | US  |
| 2 | SP  | !   | "   | #   | \$  | %   | &   | '   |
|   | (   | )   | *   | +   | ,   | -   | .   | /   |
| 3 | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7   |
|   | 8   | 9   | :   | ;   | <   | =   | >   | ?   |
| 4 | @   | A   | B   | C   | D   | E   | F   | G   |
|   | H   | I   | J   | K   | L   | M   | N   | O   |
| 5 | P   | Q   | R   | S   | T   | U   | V   | W   |
|   | X   | Y   | Z   | [   | \   | ]   | ^   | _   |
| 6 | `   | a   | b   | c   | d   | e   | f   | g   |
|   | h   | i   | j   | k   | l   | m   | n   | o   |
| 7 | p   | q   | r   | s   | t   | u   | v   | w   |
|   | x   | y   | z   | {   |     | }   | ~   | DEL |

### 17.4 • UTF-8 Input

Converting U+uvwxxyz. ASCII: 0 to 7F; Korean: AC00 to D7A3.

- 0 to 7F: 0yyyyzzzz.
- 80 to 7FF: 110xxxxxy 10yyzzzzz.
- 800 to FFFF: 1110wwwww 10xxxxxy 10yyzzzzz.
- 10000 to 10FFFF: 11110uvv vvvwww 10xxxxxy 10yyzzzzz.